

Calabi–Yau Theorem (Calabi Conjecture)

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Let (M, ω_0) be a compact Kähler manifold with $\dim_{\mathbb{C}} M = n$.

$$(\Rightarrow [\text{Ric}(\omega_0)] =: 2\pi c_1(M)_{\mathbb{R}} \in H_{\text{dR}}^2(M; \mathbb{R}))$$

Then for any smooth closed real $(1, 1)$ -form ρ satisfying

$$[\rho] = 2\pi c_1(M)_{\mathbb{R}},$$

there exists a unique Kähler metric ω such that

$$\begin{cases} \text{Ric}(\omega) = \rho, \\ [\omega] = [\omega_0] \in H_{\text{dR}}^2(M; \mathbb{R}). \end{cases}$$

Local-Coordinate Formulation

In local holomorphic coordinates $(z^1, \dots, z^n)$, write

$$\omega_0 = \sqrt{-1} \sum_{i,j=1}^n g_{i\bar{j}} dz^i \wedge d\bar{z}^j,$$

$$\omega = \sqrt{-1} \sum_{i,j=1}^n \left(g_{i\bar{j}} + \frac{\partial^2 \varphi}{\partial z^i \partial \bar{z}^j} \right) dz^i \wedge d\bar{z}^j.$$

$$(\Longleftrightarrow [\omega] = [\omega_0] \text{ by the } \partial\bar{\partial}\text{-lemma.})$$

$$\text{Ric}(\omega) = \rho \Longleftrightarrow \det \left(g_{i\bar{j}} + \frac{\partial^2 \varphi}{\partial z^i \partial \bar{z}^j} \right) = \boxed{e^F \det (g_{i\bar{j}})}$$

Note: $F \in C^\infty(M, \mathbb{R})$ is the normalized smooth function determined by ρ relative to ω_0 .

Notation	Meaning
ω_0	“Omega naught”: the reference Kähler form.
(M, ω_0)	“M equipped with the Kähler form ω_0 .”
$\dim_{\mathbb{C}} M = n$	“The complex dimension of M is n.”
$\text{Ric}(\omega_0)$	“The Ricci form of ω_0 .”
$[\text{Ric}(\omega_0)]$	“The de Rham cohomology class of the Ricci form of ω_0 .”
$c_1(M)_{\mathbb{R}}$	“The first Chern class of M with real coefficients.”
$H_{\text{dR}}^2(M; \mathbb{R})$	“The second de Rham cohomology group of M with real coefficients.”
ρ	“Rho”: the prescribed smooth, closed, real differential form of type one-one that is to be the Ricci form.
$(1, 1)$ -form	“A differential form of type one-one.”
$[\rho]$	“The de Rham cohomology class of rho.”
$[\omega] = [\omega_0]$	“The de Rham cohomology class of omega equals the de Rham cohomology class of ω_0 .”
$\frac{\partial^2 \varphi}{\partial z^i \partial \bar{z}^j}$	“The mixed second partial derivative of phi with respect to z superscript i and the complex conjugate of z superscript j.”
g	“g”: the reference Kähler metric tensor associated with ω_0 .
$g_{i\bar{j}}$	“The corresponding component of the reference Kähler metric in the chosen holomorphic coordinates.”
$(g_{i\bar{j}})$	“The matrix whose entry in row i and column j is g with subscripts i and barred j”: the Hermitian coordinate matrix of the reference Kähler metric.”
$\det(g_{i\bar{j}})$	“The determinant of the matrix whose entry in row i and column j is g with subscripts i and barred j.”
$\det\left(g_{i\bar{j}} + \frac{\partial^2 \varphi}{\partial z^i \partial \bar{z}^j}\right)$	“The determinant of the matrix whose entry in row i and column j is the sum of g with subscripts i and barred j and the mixed second partial derivative of phi with respect to z superscript i and the complex conjugate of z superscript j.”
$F \in C^\infty(M, \mathbb{R})$	“F is a smooth function from M to $\mathbb{R}$.”

Explanations

1. The cohomological condition.

We use the Ricci-form convention

$$\text{Ric}(\omega_0) = -\sqrt{-1} \partial \bar{\partial} \log \det(g_{i\bar{j}}),$$

where $(g_{i\bar{j}})$ is the coefficient matrix of ω_0 . With this convention, $[\text{Ric}(\omega_0)] = 2\pi c_1(M)_{\mathbb{R}}$, where $c_1(M)_{\mathbb{R}}$ is the image of the integral first Chern class in real de Rham cohomology. Changing the metric preserves this class, so the prescribed form ρ must be closed and represent it. The form ρ need not be positive.

2. Why the potential describes a fixed Kähler class.

The term $\sqrt{-1} \partial \bar{\partial} \varphi$ is exact, so adding it to ω_0 preserves the de Rham class. Conversely, on a compact Kähler manifold, the $\partial \bar{\partial}$ -lemma says that the difference of two cohomologous Kähler forms has this form for a globally defined smooth real function φ .

The condition $\omega > 0$ means that the Hermitian coefficient matrix in the Local-Coordinate Formulation is positive-definite at every point. It ensures that the resulting form defines a Kähler metric; a positive determinant alone is not sufficient in general. Thus the unknown metric is encoded by one real-valued potential φ , subject to positivity.

3. How F is chosen.

The function F in the Local-Coordinate Formulation is chosen from the prescribed form ρ . The cohomological hypothesis gives $[\text{Ric}(\omega_0) - \rho] = 0$, so the $\partial \bar{\partial}$ -lemma provides a smooth real function F satisfying

$$\text{Ric}(\omega_0) - \rho = \sqrt{-1} \partial \bar{\partial} F.$$

This determines F up to a constant on each connected component. Choose these constants so that

$$\int_{M_\alpha} e^F \omega_0^n = \int_{M_\alpha} \omega_0^n \quad \text{for every connected component } M_\alpha.$$

For connected M , this is a single condition integrated over M .

This normalization is necessary because cohomologous Kähler forms have the same total volume on each component:

$$\int_{M_\alpha} \omega^n = \int_{M_\alpha} \omega_0^n.$$

Once normalized, F cannot be shifted by a nonzero constant while retaining solvability of the volume equation. Adding a constant to φ leaves ω unchanged and cannot compensate for such a shift in F .

4. Why the Ricci equation becomes a determinant equation.

The local determinant ratio is the globally defined ratio of volume forms, so the change in the Ricci form can be written as

$$\text{Ric}(\omega) - \text{Ric}(\omega_0) = -\sqrt{-1} \partial \bar{\partial} \log \frac{\omega^n}{\omega_0^n}.$$

If $\text{Ric}(\omega) = \rho$, this identity and the defining equation for F imply that $\log(\omega^n/\omega_0^n) - F$ has vanishing $\partial \bar{\partial}$. On a compact connected component this function is constant; the volume normalization forces that constant to be zero. Conversely, the volume equation below immediately gives $\text{Ric}(\omega) = \rho$ by the same identity. Hence

$$\text{Ric}(\omega) = \rho \iff (\omega_0 + \sqrt{-1} \partial \bar{\partial} \varphi)^n = e^F \omega_0^n.$$

Expanding the top wedge powers gives exactly the determinant equation in the Local-Coordinate Formulation. On the positive-definite branch specified there, this complex Monge–Ampère equation is elliptic.

5. What is unique.

The Kähler metric ω is unique. Its potential φ is unique up to addition of a constant on each connected component. One may fix this freedom by imposing

$$\int_{M_\alpha} \varphi \omega_0^n = 0 \quad \text{for every connected component } M_\alpha.$$

In particular, if $c_1(M)_\mathbb{R} = 0$ and one chooses $\rho = 0$, then every Kähler class contains a unique Ricci-flat Kähler metric.

Definitions

$$S^2 = \{(X, Y, Z) \in \mathbb{R}^3 : X^2 + Y^2 + Z^2 = 1\}, \quad N = (0, 0, 1), \quad S = (0, 0, -1).$$

The sphere has the subspace topology and the outward orientation. Identify it with $\mathbf{CP}^1$ by the charts

$$z = \frac{X + iY}{1 + Z} \quad \text{on } U_0 = S^2 \setminus \{S\}, \quad w = \frac{X - iY}{1 - Z} \quad \text{on } U_\infty = S^2 \setminus \{N\}.$$

Then $w = 1/z$. In homogeneous coordinates $[S : T]$ (the homogeneous symbol S is distinct from the south pole), $z = T/S$, $w = S/T$, and $\infty = [0 : 1]$. Put $D = 1 + x^2 + y^2$ for $z = x + iy$. The inverse of the z -chart is

$$\Phi(z) = \frac{(2x, 2y, 1 - x^2 - y^2)}{D}.$$

The unit round metric and its area form are

$$g = \frac{4}{D^2}(dx^2 + dy^2), \quad \Omega = \frac{4}{D^2}dx \wedge dy = \frac{2i}{D^2}dz \wedge d\bar{z}, \quad \int_{S^2} \Omega = 4\pi.$$

Here $g_{z\bar{z}} = g_{\mathbb{C}}(\partial_z, \partial_{\bar{z}}) = 2/D^2$. The tangent-line Hermitian metric has $h_T(\partial_z, \partial_z) = 2/D^2$; its dual on K has $h_K(dz, dz) = D^2/2$.

$$A_K = \frac{2\bar{z}}{D}dz, \quad F_K = -\frac{2}{D^2}dz \wedge d\bar{z} = i\Omega, \quad \frac{i}{2\pi} \int_{\mathbf{CP}^1} F_K = -2.$$

Hermitian forms are linear in the first argument. The curvature conventions are $\Theta = \nabla^2$ and $c_1(L, h) = iF/(2\pi)$. The Gaussian curvature is the scalar $K_G = 1$; it is not the two-form F_K .

Analytic versus algebraic. Unless explicitly stated otherwise, $\mathcal{O}_X$ on $\mathbf{CP}^1$ means the analytic holomorphic-function sheaf. Thus $\mathcal{O}_X(U_0) = \mathcal{O}(\mathbb{C})$, not $\mathbb{C}[z]$. Algebraic examples explicitly say so. A skyscraper i_*A at p has sections A when $p \in U$ and zero otherwise; these examples are in abelian sheaves unless stated to be module sheaves.

A fixed triangulated sphere. Use the four unit vectors proportional to $(1, 1, 1)$, $(1, -1, -1)$, $(-1, 1, -1)$, $(-1, -1, 1)$. Radial projection identifies the boundary K of their tetrahedron with S^2 . The open star of a vertex is the locus where its barycentric coordinate is positive. Its nonempty multiple intersections are the open stars of faces. They are contractible, and the fourfold intersection is empty. These four sets give the good cover used in the homology and Čech examples. Reorder vertex labels if necessary so that the displayed fundamental cycle agrees with the outward orientation.

A fixed smooth cutoff. Let $\eta(t) = e^{-1/t}$ for $t > 0$ and $\eta(t) = 0$ otherwise. Put

$$\rho_0 = \frac{\eta(Z + 1/2)}{\eta(Z + 1/2) + \eta(1/2 - Z)}, \quad \rho_\infty = 1 - \rho_0.$$

The denominator is positive, $\text{supp } \rho_0 \subset \{Z \geq -1/2\} \subset U_0$, and $\text{supp } \rho_\infty \subset \{Z \leq 1/2\} \subset U_\infty$.

1. Topological Space

A *topological space* is a pair $(X, \mathcal{T})$, where X is a set and $\mathcal{T} \subseteq \mathcal{P}(X)$ satisfies:

- (a) $\emptyset, X \in \mathcal{T}$;
- (b) any arbitrary union of members of $\mathcal{T}$ belongs to $\mathcal{T}$;

1. Topological Space (continued)

(c) any finite intersection of members of $\mathcal{T}$ belongs to $\mathcal{T}$.

The members of $\mathcal{T}$ are the *open sets*.

Example.

Let $S^2 = \{(X, Y, Z) \in \mathbb{R}^3 : X^2 + Y^2 + Z^2 = 1\}$. Its standard topology is $\mathcal{T} = \{S^2 \cap V : V \subset \mathbb{R}^3 \text{ open}\}$. Thus $\{Z > 0\} \cap S^2$ is open, while the equator is closed. Arbitrary unions and finite intersections remain intersections with ambient open sets.

2. Metric and Metric Topology

Let X be a set. A *metric* on X is a function

$$d: X \times X \rightarrow \mathbb{R}$$

such that, for all $x, y, z \in X$,

- (a) $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$;
- (b) $d(x, y) = d(y, x)$;
- (c) $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (X, d) is called a *metric space*.

For $p \in X$ and $\varepsilon > 0$, the *open ball* centered at p with radius ε is

$$B_\varepsilon(p) = \{x \in X : d(x, p) < \varepsilon\}.$$

A subset $U \subseteq X$ is called *open* if, for every $p \in U$, there exists $\varepsilon > 0$ such that

$$B_\varepsilon(p) \subseteq U.$$

The collection

$$\mathcal{T}_d = \{U \subseteq X : \text{for every } p \in U, \text{ there exists } \varepsilon > 0 \text{ such that } B_\varepsilon(p) \subseteq U\}$$

is a topology on X , called the *metric topology induced by d* .

Example.

For $p, q \in S^2$, the chordal metric is $d(p, q) = \|p - q\|_{\mathbb{R}^3}$. Its balls are $B_r^{\mathbb{R}^3}(p) \cap S^2$, so its metric topology is exactly the subspace topology. For the poles N, S, $d(N, S) = 2$. The intrinsic round distance is instead $d_g(p, q) = \arccos(p \cdot q)$, with $d_g(N, S) = \pi$; it induces the same topology, not the same numerical distance.

3. Compactness

A topological space X is *compact* if every open cover $X = \bigcup_{\alpha \in A} U_\alpha$ has a finite subcover:

$$X = U_{\alpha_1} \cup \cdots \cup U_{\alpha_N}.$$

Example.

The subset $S^2 \subset \mathbb{R}^3$ is closed, as the inverse image of $\{1\}$ under $X^2 + Y^2 + Z^2$, and bounded. Heine–Borel therefore makes it compact. In particular every relative open cover, not only the two stereographic charts, has a finite subcover.

4. Paracompactness

A topological space X is *paracompact* if every open cover of X admits a locally finite open refinement. An open cover $\{V_\beta\}_{\beta \in B}$ *refines* an open cover $\{U_\alpha\}_{\alpha \in A}$ if for every $\beta \in B$, there exists $\alpha \in A$ such that $V_\beta \subseteq U_\alpha$.

The collection $\{V_\beta\}_{\beta \in B}$ is *locally finite* if every point $x \in X$ has a neighborhood that intersects only finitely many of the sets V_β . Some authors include Hausdorffness in the definition of paracompactness. In our setting, manifolds are assumed to be Hausdorff.

Example.

The unit sphere is compact because it is closed and bounded in $\mathbb{R}^3$ (Heine–Borel). Every open cover therefore has a finite subcover, which is itself a locally finite open refinement. This verifies paracompactness of this particular S^2 .

5. Partition of unity

Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of a smooth manifold M . A *smooth partition of unity subordinate to this cover* is a family $\{\varphi_\alpha\}_{\alpha \in A}$ of smooth functions $\varphi_\alpha : M \rightarrow [0, 1]$ such that

$$\text{supp } \varphi_\alpha \subseteq U_\alpha, \quad \sum_{\alpha \in A} \varphi_\alpha(x) = 1 \quad (x \in M),$$

and the family $\{\text{supp } \varphi_\alpha\}$ is locally finite. Here

$$\text{supp } \varphi_\alpha = \overline{\{x \in M : \varphi_\alpha(x) \neq 0\}}.$$

Local finiteness ensures that the sum is locally a finite sum of smooth functions. Zero functions are allowed.

Example.

On S^2 , put $\rho_0 = \eta(Z + 1/2)/(\eta(Z + 1/2) + \eta(1/2 - Z))$ and $\rho_\infty = 1 - \rho_0$. The denominator is everywhere positive. These smooth functions sum to 1, with supports in $\{Z \geq -1/2\} \subset U_0$ and $\{Z \leq 1/2\} \subset U_\infty$. This is an explicit partition subordinate to the two standard charts.

6. Topological manifold

An m -dimensional *topological manifold* is a Hausdorff, second-countable topological space M in which each point has an open neighborhood homeomorphic to an open subset of $\mathbb{R}^m$. Such a homeomorphism

$$\varphi : U \longrightarrow \varphi(U) \subseteq \mathbb{R}^m$$

defines a *coordinate chart* (U, φ) . Hausdorff means that any two distinct points have disjoint open neighborhoods; second-countable means that the topology has a countable basis.

Example.

On $S^2 \setminus \{S\}$, $z = (X + iY)/(1 + Z)$ is a homeomorphism onto $\mathbb{C} \simeq \mathbb{R}^2$ with inverse $\Phi(x + iy) = (2x, 2y, 1 - x^2 - y^2)/(1 + x^2 + y^2)$. Together with the w-chart at S it verifies local Euclidean dimension two. The subspace topology is Hausdorff and second countable.

7. Smooth manifold

A *smooth manifold* is a topological manifold equipped with a smooth structure. A smooth structure can be specified by an atlas of charts $\{(U_\alpha, \varphi_\alpha)\}$ covering M , whose transition maps

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \longrightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

are C^∞ . Equivalently, one specifies a maximal smooth atlas. Two smooth atlases define the same smooth structure if their union is a smooth atlas. Thus an atlas need not itself be maximal to specify a smooth manifold.

Example.

The fixed charts $z = x + iy$ and $w = 1/z$ identify the underlying S^2 with a smooth real surface. On the overlap, $(x, y) \mapsto (x/(x^2 + y^2), -y/(x^2 + y^2))$ is smooth with smooth inverse. The Hausdorff and countable-basis conditions follow from the subspace topology in $\mathbb{R}^3$.

8. Riemannian manifold

A *Riemannian manifold* is a pair (M, g) consisting of a smooth manifold and a Riemannian metric.

Example.

The pair (S^2, g) with $g_p(u, v) = u \cdot v$ for $u, v \in T_p S^2 = p^\perp$ is a Riemannian manifold. In the fixed z-chart, $g = 4(1 + |z|^2)^{-2}(dx^2 + dy^2)$. Here juxtaposition denotes symmetric metric tensors, not wedge products.

8.1. Riemannian metric

A Riemannian metric is a smooth section g of $\text{Sym}^2(T^*M)$ such that at every point

$$g_p : T_p M \times T_p M \rightarrow \mathbb{R}$$

is symmetric, bilinear, and positive-definite. In particular,

$$g_p(X, Y) = g_p(Y, X), \quad g_p(X, X) > 0 \quad \text{for } X \neq 0.$$

Example.

On S^2 , restrict the Euclidean dot product to each tangent plane. It is symmetric and bilinear, and $g_p(u, u) = \|u\|^2 > 0$ for $u \neq 0$. Its coordinate matrix in the z-chart is $4(1 + |z|^2)^{-2}I_2$, a smooth positive-definite matrix. This checks the smooth family condition as well as pointwise positivity.

8.2. Levi-Civita connection

A connection on TM is an $\mathbb{R}$ -bilinear operation $(X, Y) \mapsto \nabla_X Y$ on smooth vector fields satisfying

$$\nabla_{fX} Y = f \nabla_X Y, \quad \nabla_X (fY) = X(f)Y + f \nabla_X Y.$$

The *Levi-Civita connection* is the unique connection with

$$\begin{aligned} \nabla_X Y - \nabla_Y X &= [X, Y], & (\text{torsion-free}) \\ X(g(Y, Z)) &= g(\nabla_X Y, Z) + g(Y, \nabla_X Z), & (\text{metric compatible}) \end{aligned}$$

where $[X, Y]$ is the Lie bracket, defined by $[X, Y](f) = X(Y(f)) - Y(X(f))$.

Example.

For tangent vector fields U, V on the unit sphere, $\nabla_U V = P_p(D_U V) = D_U V + \langle U, V \rangle p$, where $P_p(a) = a - \langle a, p \rangle p$ and D is the ambient Euclidean derivative. Differentiating $\langle V, p \rangle = 0$ gives $\langle D_U V, p \rangle = -\langle V, U \rangle$. Projection makes the output tangent, and the Euclidean product rule verifies metric compatibility. The normal correction is symmetric in U, V , so torsion is zero.

8.3. Real tangent space and tangent bundle

A tangent vector at p can be defined as an $\mathbb{R}$ -linear map $v : C^\infty(M, \mathbb{R}) \rightarrow \mathbb{R}$ satisfying

$$v(fh) = f(p)v(h) + h(p)v(f).$$

These derivations form the real vector space $T_p M$ of dimension m . The tangent bundle is

$$TM = \bigsqcup_{p \in M} T_p M, \quad \pi : TM \rightarrow M, \quad v \in T_p M \mapsto p,$$

with the smooth vector-bundle structure induced by coordinate charts.

Example.

At $p \in S^2$, $T_p S^2 = p^\perp$. If $v \in p^\perp$ and $\tilde{f}$ extends a smooth sphere function near p , the derivation is $v(f) = D\tilde{f}_p(v)$, independent of the extension. At N , $v = e_1$ takes the coordinate restrictions X, Y, Z to $1, 0, 0$, respectively. The total tangent bundle consists of pairs (p, v) with $\|p\| = 1$ and $p \cdot v = 0$.

9. Complex manifold and complex structure

A *complex manifold* of complex dimension n is a Hausdorff, second-countable space with an atlas of homeomorphisms

$$\varphi_\alpha : U_\alpha \longrightarrow \varphi_\alpha(U_\alpha) \subseteq \mathbb{C}^n$$

onto open sets, covering M , with holomorphic transition maps. The underlying smooth manifold has real dimension $2n$.

An *almost complex structure* is a smooth real vector-bundle endomorphism $J : TM \rightarrow TM$ with $J^2 = -\text{Id}$. It is *integrable* if there is a complex atlas inducing J ; then it is called a *complex structure*. In holomorphic coordinates $z^j = x^j + iy^j$,

$$J \frac{\partial}{\partial x^j} = \frac{\partial}{\partial y^j}, \quad J \frac{\partial}{\partial y^j} = -\frac{\partial}{\partial x^j}.$$

Example.

On the oriented unit sphere put $J_p v = p \times v$ for $v \perp p$. The vector triple-product identity gives $J_p^2 v = -v$. In our z -chart, $J \partial_x = \partial_y$, and the second chart satisfies $w = 1/\bar{z}$. Thus this J is induced by a complex atlas, not merely an almost complex structure.

10. Tensor product

For vector spaces V, W over a field $\mathbb{F}$, their tensor product is a vector space $V \otimes_{\mathbb{F}} W$ with a bilinear map $(v, w) \mapsto v \otimes w$ such that every bilinear map $B : V \times W \rightarrow U$ into an $\mathbb{F}$ -vector space factors uniquely as

$$B(v, w) = \tilde{B}(v \otimes w)$$

for a linear map $\tilde{B} : V \otimes_{\mathbb{F}} W \rightarrow U$. This universal property determines the tensor product up to a unique isomorphism respecting the bilinear map.

Example.

At $N \in S^2$, take the tangent basis e_1, e_2 and dual basis ϵ^1, ϵ^2 . Then $(\epsilon^1 \otimes \epsilon^2)(e_1, e_2) = 1$, while $(\epsilon^1 \wedge \epsilon^2)(e_1, e_2) = 1$ and its value on (e_2, e_1) is -1 . The first tensor is not alternating, and the dual covector satisfies $\epsilon^1(e_1) = 1$.

11. Complexified tangent bundle

The *complexified tangent bundle* is

$$T_{\mathbb{C}}M = TM \otimes_{\mathbb{R}} \mathbb{C}, \quad (T_{\mathbb{C}}M)_p = T_pM \otimes_{\mathbb{R}} \mathbb{C}.$$

If M has real dimension m , its complexification has complex rank m . An almost complex structure extends $\mathbb{C}$ -linearly to $T_{\mathbb{C}}M$ and gives

$$T_{\mathbb{C}}M = T^{1,0}M \oplus T^{0,1}M.$$

When $\dim_{\mathbb{C}} M = n$, each summand has complex rank n , whereas $T_{\mathbb{C}}M$ has complex rank $2n$.

Example.

11. Complexified tangent bundle (continued)

At N on S^2 , $T_N S^2 = \mathbb{R}e_1 \oplus \mathbb{R}e_2$, so its complexification is $\mathbb{C}e_1 \oplus \mathbb{C}e_2$, of complex dimension two. The extended J has eigenvectors $e_1 - ie_2$ and $e_1 + ie_2$, with eigenvalues i and -i. These are the one-dimensional (1,0) and (0,1) spaces. Complexification and taking the holomorphic tangent line are different operations.

12. Holomorphic tangent bundle

For a complex manifold (M, J) , define the eigenbundles

$$T^{1,0}M = \ker(J - i \text{Id}), \quad T^{0,1}M = \ker(J + i \text{Id})$$

inside $T_{\mathbb{C}}M$. In holomorphic coordinates,

$$\begin{aligned} \frac{\partial}{\partial z^j} &= \frac{1}{2} \left(\frac{\partial}{\partial x^j} - i \frac{\partial}{\partial y^j} \right), \\ \frac{\partial}{\partial \bar{z}^j} &= \frac{1}{2} \left(\frac{\partial}{\partial x^j} + i \frac{\partial}{\partial y^j} \right) \end{aligned}$$

form local frames for $T^{1,0}M$ and $T^{0,1}M$, respectively. The holomorphic coordinate frames give $T^{1,0}M$ its natural *holomorphic vector-bundle structure*. A smooth section $\sum_j a^j \partial/\partial z^j$ is a holomorphic section precisely when the functions a^j are holomorphic. Thus not every smooth section of the holomorphic tangent bundle is holomorphic.

Example.

On $\mathbf{CP}^1$, $\partial_w = -z^2 \partial_z$ and $dw = -z^{-2} dz$. Pairing gives $dw(\partial_w) = 1$ on the overlap. Thus the tangent and cotangent transition functions are reciprocal, and each bundle has complex rank one.

13. Chern connection

A holomorphic vector bundle $E \rightarrow M$ has local trivializations $E|_U \simeq U \times \mathbb{C}^r$ with holomorphic, fiberwise linear transition maps. A *Hermitian metric* h on E is a smooth family of positive-definite Hermitian inner products on its fibers. We take h to be complex-linear in its first argument and conjugate-linear in its second.

The *Chern connection* is the unique complex connection

$$\nabla : \Gamma(E) \longrightarrow \Omega^1(M; E)$$

such that $\nabla h = 0$ and $\nabla^{0,1} = \bar{\partial}_E$. Here $\Gamma(E)$ denotes smooth sections, $\Omega^1(M; E)$ denotes E -valued one-forms, and $\bar{\partial}_E$ applies $\bar{\partial}$ to the coefficient functions in a local holomorphic frame. A complex connection satisfies $\nabla(fs) = df \otimes s + f \nabla s$. For real vector fields X , compatibility with h means

$$X(h(s, t)) = h(\nabla_X s, t) + h(s, \nabla_X t).$$

We use the curvature convention

$$\Theta = \nabla^2, \quad \Theta(X, Y) = \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X, Y]}.$$

The curvature of a Chern connection is of type $(1, 1)$.

For a holomorphic line bundle L with Hermitian metric h , the Chern connection is the unique connection with $\nabla^{0,1} = \bar{\partial}_L$ which is metric-compatible:

$$Vh(s, t) = h(\nabla_V s, t) + h(s, \nabla_V t)$$

13. Chern connection (continued)

for every real vector field V and smooth sections s, t . In a holomorphic frame e , with $h_e = h(e, e)$, its connection form and curvature are

$$A_e = \partial \log h_e, \quad F_{\nabla} = dA_e = -\partial \bar{\partial} \log h_e.$$

Example.

The Hermitian metric $h_z = (1 + |z|^2)^2/2$ on $K_{\mathbf{CP}^1}$ gives the Chern connection $A_z = \partial \log h_z = 2\bar{z}(1 + |z|^2)^{-1}dz$. Its curvature is $F_K = -2(1 + |z|^2)^{-2}dz \wedge d\bar{z}$, a global $(1,1)$ -form. The factor $1/2$ in h disappears when differentiating its logarithm.

14. Symplectic form

A *symplectic form* is a real differential two-form $\omega \in \Omega^2(M; \mathbb{R})$ such that $d\omega = 0$ and ω is nondegenerate:

$$\omega_p(X, Y) = 0 \text{ for every } Y \in T_p M \implies X = 0.$$

A pair (M, ω) is then a *symplectic manifold*.

Example.

On S^2 , $\Omega_p(u, v) = p \cdot (u \times v)$ is closed for dimension reasons. If $u \neq 0$, choose $v = J_p u = p \times u$; then $\Omega_p(u, J_p u) = \|u\|^2 > 0$. Hence it is nondegenerate and gives a symplectic form.

15. Compatibility, Hermitian metrics, and Kähler manifolds

A Riemannian metric g on an almost complex manifold (M, J) is *J-compatible* if

$$g(JX, JY) = g(X, Y).$$

Its associated real two-form, with our convention, is

$$\omega(X, Y) = g(JX, Y).$$

It is alternating because $g(JX, Y) = -g(X, JY)$, and

$$\omega(X, JX) = g(X, X) > 0 \quad (X \neq 0).$$

If J is integrable, g is called a *Hermitian metric* on (M, J) . If, additionally, $d\omega = 0$, the tuple (M, J, g, ω) is a *Kähler manifold*, and ω is its *Kähler form*. Equivalently, on a complex manifold a Kähler form is a closed positive real $(1,1)$ -form; see Definition 20.

Example.

For the unit sphere, take $g_p(u, v) = u \cdot v$ and $J_p u = p \times u$. Then $g(Ju, Jv) = g(u, v)$ and $g(Ju, v) = \Omega(u, v)$. The chosen complex atlas makes J integrable, while $d\Omega = 0$, so this gives a Kähler structure on $\mathbf{CP}^1$. In the z -chart, $\Omega = 2i(1 + |z|^2)^{-2}dz \wedge d\bar{z}$ and $g_{z\bar{z}} = 2(1 + |z|^2)^{-2}$.

16. Riemann curvature tensor

For the Levi-Civita connection we define

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

This is a tensor measuring the failure of covariant derivatives to commute, with the Lie-bracket correction included.

Example.

For the unit round S^2 with the stated sign convention, $R(U, V)W = \langle V, W \rangle U - \langle U, W \rangle V$. For an oriented orthonormal pair e_1, e_2 at p , $R(e_1, e_2)e_2 = e_1$, and the Gaussian curvature is $g(R(e_1, e_2)e_2, e_1) = 1$. This is tangent-bundle curvature, not the canonical-line curvature form.

17. Ricci curvature tensor

The *Ricci tensor* is the contraction

$$\text{Ric}_g(X, Y) = \text{tr}(Z \mapsto R(Z, X)Y).$$

If $\{e_1, \dots, e_m\}$ is a real g -orthonormal basis of $T_p M$, then

$$\begin{aligned} \text{Ric}_g(X, Y) &= \sum_{a=1}^m g(R(e_a, X)Y, e_a), \\ m &= \dim_{\mathbb{R}} M. \end{aligned}$$

In particular, this sum has $2n$ terms on a complex n -dimensional manifold. The Ricci tensor is a symmetric real $(0, 2)$ -tensor; the Ricci form in Definition 21 is an alternating two-form.

Example.

On the round S^2 , contract the preceding R with an orthonormal pair. The term for e_1 in $\text{Ric}_g(e_1, e_1)$ is zero; the term for e_2 is $g(R(e_2, e_1)e_1, e_2) = 1$. Likewise $\text{Ric}_g(e_2, e_2) = 1$ and the mixed term vanishes, so $\text{Ric}_g = g$.

18. Differential forms

A real differential k -form is a smooth section of $\Lambda^k T^*M$:

$$\Omega^k(M; \mathbb{R}) = \Gamma(\Lambda^k T^*M).$$

At p , its value is an alternating k -linear map $(T_p M)^k \rightarrow \mathbb{R}$. Complex-valued forms instead lie in

$$\Omega^k(M; \mathbb{C}) = \Gamma(\Lambda^k T^*M \otimes_{\mathbb{R}} \mathbb{C}).$$

We abbreviate $\Omega^k(M; \mathbb{R})$ to $\Omega^k(M)$ when there is no ambiguity. The wedge product $\alpha \wedge \beta$ is the exterior product of forms; it has degree $k + \ell$ when α has degree k and β degree ℓ .

For $q \geq 1$, a smooth real differential q -form on an n -manifold M is a family $\omega = (\omega_p)_{p \in M}$ of alternating real q -linear maps $(T_p M)^q \rightarrow \mathbb{R}$ with the following smoothness property. For every smooth chart $\varphi : U \rightarrow V \subseteq \mathbb{R}^n$, put

$$E_i(p) = \left. \frac{d}{dt} \right|_{t=0} \varphi^{-1}(\varphi(p) + te_i).$$

For every $i_1, \dots, i_q$, the function

$$V \longrightarrow \mathbb{R}, \quad y \longmapsto \omega_{\varphi^{-1}(y)}(E_{i_1}(\varphi^{-1}(y)), \dots, E_{i_q}(\varphi^{-1}(y)))$$

must be C^∞ . A smooth 0-form is a function $f : M \rightarrow \mathbb{R}$ with $f \circ \varphi^{-1}$ smooth in every chart. The vector space of smooth q -forms is $\Omega^q(M)$.

Example.

The area form of the unit sphere is $\Omega_p(u, v) = p \cdot (u \times v)$ for $u, v \in T_p S^2$. This is alternating and depends smoothly on p . In the z -chart it is $\Omega = 4(1 + |z|^2)^{-2} dx \wedge dy$, so its coordinate coefficient is explicitly smooth.

On the z -chart of $\mathbf{CP}^1$, $dz = dx + i dy$ is a complex-valued real-linear one-form. It takes ∂_x to 1 and ∂_y to i . The global form $i\Omega$ is a complex-valued two-form on all of S^2 ; its imaginary part is Ω .

19. Exterior derivative and de Rham cohomology

The exterior derivative consists of $\mathbb{R}$ -linear maps $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ characterized by

$$\begin{aligned} df(X) &= X(f), \\ d^2 &= 0, \\ d(\alpha \wedge \beta) &= d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta, \\ \alpha &\in \Omega^k(M). \end{aligned}$$

It extends complex-linearly to complex-valued forms. A form α is *closed* if $d\alpha = 0$ and *exact* if $\alpha = d\beta$ for some form β of one lower degree. The de Rham cohomology is

$$H_{\text{dR}}^k(M; \mathbb{R}) = \frac{\ker(d : \Omega^k(M) \rightarrow \Omega^{k+1}(M))}{\text{im}(d : \Omega^{k-1}(M) \rightarrow \Omega^k(M))}.$$

19. Exterior derivative and de Rham cohomology (continued)

Set $\Omega^{-1}(M) = 0$ for the case $k = 0$. For a closed form α , its cohomology class is denoted $[\alpha]$. Closed forms α, β have the same class exactly when $\alpha - \beta$ is exact.

The exterior derivative $d : \Omega^q(M) \rightarrow \Omega^{q+1}(M)$ is the real-linear operator defined in a chart $\varphi = (x^1, \dots, x^n) : U \rightarrow V$ by

$$\omega|_U = \sum_I (a_I \circ \varphi) dx^I, \quad (d\omega)|_U = \sum_{j,I} \left(\frac{\partial a_I}{\partial y_j} \circ \varphi \right) dx^j \wedge dx^I.$$

Here $I = (i_1 < \dots < i_q)$, $dx^I = dx^{i_1} \wedge \dots \wedge dx^{i_q}$, y denotes coordinates on V , and $a_I \in C^\infty(V, \mathbb{R})$. For $q = 0$, use the empty index with $dx^\varnothing = 1$. These local expressions agree on overlaps.

Example.

Integration induces $H_{\text{dR}}^2(S^2; \mathbb{R}) \simeq \mathbb{R}$, $[\alpha] \mapsto \int_{S^2} \alpha$. The class $[\Omega/(4\pi)]$ maps to 1. In degrees 0 and 1 the groups are $\mathbb{R}$ and 0, respectively. These are the computed groups of this sphere, not part of the general definition. The global height function on S^2 is $H = (1 - x^2 - y^2)/(1 + x^2 + y^2)$ in the z -chart. Its differential is $dH = -4(x dx + y dy)/(1 + x^2 + y^2)^2$. Applying d again gives zero: the mixed partials cancel in the coefficient of $dx \wedge dy$.

20. Forms of type (p, q) and $d = \partial + \bar{\partial}$

On a complex manifold, forms of type (p, q) are locally sums of terms

$$a_{I\bar{J}}(z) dz^{i_1} \wedge \dots \wedge dz^{i_p} \wedge d\bar{z}^{j_1} \wedge \dots \wedge d\bar{z}^{j_q},$$

with smooth complex-valued coefficients. Their space is $\Omega^{p,q}(M)$, and

$$\Omega^k(M; \mathbb{C}) = \bigoplus_{p+q=k} \Omega^{p,q}(M).$$

The exterior derivative decomposes as $d = \partial + \bar{\partial}$, where

$$\partial : \Omega^{p,q} \rightarrow \Omega^{p+1,q}, \quad \bar{\partial} : \Omega^{p,q} \rightarrow \Omega^{p,q+1}.$$

For a function f , the operators are

$$\partial f = \sum_j \frac{\partial f}{\partial z^j} dz^j, \quad \bar{\partial} f = \sum_j \frac{\partial f}{\partial \bar{z}^j} d\bar{z}^j.$$

They satisfy $\partial^2 = \bar{\partial}^2 = 0$ and $\partial\bar{\partial} + \bar{\partial}\partial = 0$. The notation $\partial\bar{\partial}f$ means $\partial(\bar{\partial}f)$. This two-term decomposition of d uses integrability; an arbitrary almost complex structure need not have it.

Example.

On the z -chart of $\mathbf{CP}^1$, take $f = z\bar{z}$. Then $\partial f = \bar{z} dz$, $\bar{\partial} f = z d\bar{z}$, and $\partial\bar{\partial}f = dz \wedge d\bar{z}$. These have types $(1,0)$, $(0,1)$, $(1,1)$ respectively. The function f is used only on this chart, not as a global function on the sphere.

21. Ricci real (1,1)-form

For a Kähler metric, let $g_{\mathbb{C}}$ be the complex-bilinear extension of the real metric and set

$$g_{i\bar{j}} = g_{\mathbb{C}} \left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial \bar{z}^j} \right).$$

Then, consistently with Definition 15,

$$\omega = i \sum_{i,j=1}^n g_{i\bar{j}} dz^i \wedge d\bar{z}^j.$$

The *Ricci form* is

$$\rho_{\omega} = -i\partial\bar{\partial} \log \det(g_{i\bar{j}}).$$

Equivalently,

$$\rho_{\omega} = -i \sum_{i,j=1}^n \frac{\partial^2 \log \det(g_{k\bar{\ell}})}{\partial z^i \partial \bar{z}^j} dz^i \wedge d\bar{z}^j,$$

$$\rho_{\omega}(X, Y) = \text{Ric}_g(JX, Y).$$

These local expressions define a global closed real (1,1)-form. The Ricci form is not necessarily positive. Some authors denote this form by $\text{Ric}(\omega)$; that notation refers to the two-form, whereas Ric_g above denotes the symmetric tensor.

Example.

For the round metric on $\mathbf{CP}^1$, $g_{z\bar{z}} = 2(1 + |z|^2)^{-2}$. Hence $\rho = -i\partial\bar{\partial} \log g_{z\bar{z}} = 2i(1 + |z|^2)^{-2} dz \wedge d\bar{z} = \Omega$. Its integral is 4π , and $\rho(U, V) = \text{Ric}_g(JU, V) = g(JU, V)$. The tensor $\text{Ric}_g = g$ is symmetric, whereas $\rho = \Omega$ is alternating.

22. First Chern class

For a smooth complex vector bundle $E \rightarrow M$, the first Chern class is an integral characteristic class

$$c_1(E) \in H^2(M; \mathbb{Z}), \quad c_1(E) = c_1(\det E).$$

Write $c_1(E)_{\mathbb{R}}$ for its image under the natural map to $H^2_{\text{dR}}(M; \mathbb{R})$. Choose a Hermitian metric on E and a connection compatible with that metric. With $\Theta = \nabla^2$ as in Definition 13, Chern–Weil theory gives

$$c_1(E)_{\mathbb{R}} = \left[\frac{i}{2\pi} \text{Tr } \Theta \right].$$

The displayed form is real and closed, and its cohomology class is independent of the chosen Hermitian metric and compatible connection. If E is holomorphic, one may use its Chern connection; then the form also has type $(1, 1)$. Requiring only that a connection be complex-linear would not guarantee that this particular representative is a real-valued form. The curvature formula determines the real image of $c_1(E)$; it does not detect any torsion in the integral class. See [1].

For a complex manifold, $c_1(M) := c_1(T^{1,0}M)$. On a Kähler manifold,

$$\left[\frac{\rho_{\omega}}{2\pi} \right] = c_1(M)_{\mathbb{R}}, \quad [\rho_{\omega}] = 2\pi c_1(M)_{\mathbb{R}}.$$

These equalities use real de Rham cohomology. With the definition of ρ_{ω} in Definition 21, the factor 2π must be retained. If one defines a different, normalized form $\widehat{\rho}_{\omega} = \rho_{\omega}/(2\pi)$, then $[\widehat{\rho}_{\omega}] = c_1(M)_{\mathbb{R}}$.

Example.

Let u be the integral generator of $H^2(\mathbf{CP}^1; \mathbb{Z})$ evaluating to 1 on the complex-oriented sphere. Then $c_1(T^{1,0}\mathbf{CP}^1) = 2u$ and $c_1(K_{\mathbf{CP}^1}) = -2u$. Their real images are represented by $\Omega/(2\pi)$ and $-\Omega/(2\pi)$, respectively. Here there is no torsion in H^2 , so the integral class is determined by this integer pairing; this is special to this example.

Spaces and topology

23. Open cover

An open cover of a topological space X is an indexed family $(U_i)_{i \in I}$ of open subsets of X such that $X = \bigcup_{i \in I} U_i$.

Prerequisites: Definition 1.

Example.

The two open sets $U_0 = S^2 \setminus \{S\}$ and $U_{\infty} = S^2 \setminus \{N\}$ cover S^2 . Under the fixed identification with $\mathbf{CP}^1$, these are precisely the z - and w -charts, with overlap $\mathbb{C}^{\times}$ and $w = 1/z$.

24. Open neighborhood

An open neighbourhood of a point x in a topological space X is an open subset $U \subseteq X$ containing x .

Prerequisites: Definition 1.

Example.

At $N = (0, 0, 1) \in S^2$, the cap $U = S^2 \cap \{Z > 1/2\}$ is an open neighbourhood: $N \in U$, and $\{Z > 1/2\}$ is open in $\mathbb{R}^3$. The equator is not a neighbourhood of N .

25. Continuous map

A map $f : X \rightarrow Y$ of topological spaces is continuous if $f^{-1}(V)$ is open in X for every open subset $V \subseteq Y$. A homeomorphism is a continuous bijection whose inverse is continuous.

Prerequisites: Definition 1.

Example.

The height function $H : S^2 \rightarrow \mathbb{R}$, $H(X, Y, Z) = Z$, is continuous. For example $H^{-1}((0, 2)) = S^2 \cap \{Z > 0\}$ is open. More generally every inverse image of an open interval is an ambient open slab intersected with S^2 .

26. Hausdorff space

A topological space X is Hausdorff if, for every pair of distinct points $x, y \in X$, there exist disjoint open subsets $U, V \subseteq X$ with $x \in U$ and $y \in V$.

Prerequisites: Definition 24.

Example.

For distinct $p, q \in S^2$, put $r = \|p - q\|/3$. The relative open balls $B_r^{\mathbb{R}^3}(p) \cap S^2$ and $B_r^{\mathbb{R}^3}(q) \cap S^2$ are disjoint neighbourhoods, by the triangle inequality. Thus the subspace topology of S^2 is Hausdorff.

27. Basis for a topology

A basis for a topology on X is a collection $\mathcal{B}$ of open subsets such that every open subset of X is a union of members of $\mathcal{B}$. The space is second countable if it has a countable basis.

Prerequisites: Definition 1.

Example.

The nonempty sets $S^2 \cap B_r(a)$, with $a \in \mathbb{Q}^3$ and $r \in \mathbb{Q}_{>0}$, form a countable basis. Given $p \in U$ open in S^2 , choose a small ambient ball around p inside the ambient open set defining U , then a rational-centred, rational-radius ball containing p inside it.

28. Closed set

A subset A of a topological space X is closed if its complement $X \setminus A$ is open in X .

Prerequisites: Definition 1.

Example.

The equator $E = S^2 \cap \{Z = 0\}$ is closed in S^2 : its complement is the union of the two open hemispheres $\{Z > 0\}$ and $\{Z < 0\}$, intersected with S^2 .

29. Closure

The closure of $A \subseteq X$ is $\overline{A} = \bigcap \{F \subseteq X : A \subseteq F, F \text{ closed in } X\}$. Equivalently, $x \in \overline{A}$ if every open neighbourhood of x meets A .

Prerequisites: Definition 28.

Example.

For the open northern hemisphere $H = \{p \in S^2 : Z(p) > 0\}$, its closure in S^2 is $\overline{H} = \{Z \geq 0\}$. Every neighbourhood of an equatorial point meets H ; a point with $Z < 0$ has a small ambient ball disjoint from H .

30. Locally finite family

An indexed family $(A_i)_{i \in I}$ of subsets of a topological space X is locally finite if every $x \in X$ has an open neighbourhood V for which $\{i \in I : V \cap A_i \neq \emptyset\}$ is finite.

Prerequisites: Definition 24.

Example.

The family $\{S^2 \setminus \{S\}, S^2 \setminus \{N\}\}$ is locally finite: every neighbourhood meets at most two members. In contrast $V_n = \{p \in S^2 : Z(p) > -1/n\}$ is not locally finite at N , since every V_n contains N .

31. Refinement of an open cover

An open cover $(V_a)_{a \in A}$ of X refines an open cover $(U_i)_{i \in I}$ if for every $a \in A$ there is an $i \in I$ such that $V_a \subseteq U_i$. Both families must cover X .

Prerequisites: Definition 23.

Example.

For the standard two-chart cover, $V_0 = \{Z > -3/4\}$ and $V_\infty = \{Z < 3/4\}$ form an open refinement: $V_0 \subset U_0$, $V_\infty \subset U_\infty$, and $V_0 \cup V_\infty = S^2$. All sets here are relative subsets of S^2 .

Algebra and categories

32. Abelian group

An abelian group is a set A with a binary operation $+$ such that, for all $a, b, c \in A$, $(a+b)+c = a+(b+c)$ and $a+b = b+a$, together with an element $0 \in A$ satisfying $a+0 = a$ and, for each $a \in A$, an element $-a \in A$ satisfying $a+(-a) = 0$.

Example.

The smooth real functions $C^\infty(S^2, \mathbb{R})$ form an abelian group under pointwise addition. With $H(p) = Z(p)$, the additive inverse of H is $-H$, and $H+(-H) = 0$ is the zero function on the whole sphere.

33. Ring (commutative, with identity)

A commutative ring with identity is a set R with addition and multiplication such that $(R, +)$ is an abelian group, multiplication is associative and commutative, there is 1_R with $1_R a = a 1_R = a$ for every $a \in R$, and multiplication distributes over addition in both variables.

Prerequisites: Definition 32.

Example.

On the chart $U_0 \subset \mathbf{CP}^1$, $\mathcal{O}(U_0) = \mathcal{O}(\mathbb{C})$ is a unital commutative ring under pointwise operations. It contains both z and e^z ; the latter shows why this analytic ring is larger than $\mathbb{C}[z]$.

34. Module

Let R be a commutative ring with identity. An R -module is an abelian group M with an action $R \times M \rightarrow M$, $(r, m) \mapsto rm$, satisfying, for all $r, s \in R$ and $m, n \in M$,

$$r(m+n) = rm + rn, \quad (r+s)m = rm + sm, \quad (rs)m = r(sm), \quad 1_R m = m.$$

An R -module homomorphism is an additive map preserving this action.

Prerequisites: Definition 32, Definition 33.

Example.

On $U_0 \subset \mathbf{CP}^1$, the holomorphic one-forms $\mathcal{K}(U_0) = \mathcal{O}(\mathbb{C}) dz$ form a module over $\mathcal{O}(\mathbb{C})$. For instance $(1+z)(z dz) = (z+z^2) dz$. The unit function acts as the identity.

35. Ideal

An ideal J of a commutative ring R is an additive subgroup $J \subseteq R$ such that $ra \in J$ for all $r \in R$ and $a \in J$.

Prerequisites: Definition 33.

Example.

35. Ideal (continued)

In the analytic local ring $\mathcal{O}_{\mathbf{CP}^1,0} = \mathbb{C}\{z\}$, the set $(z) = \{zf(z) : f \in \mathbb{C}\{z\}\}$ is an ideal. It contains $z + z^2$ and is closed under multiplication by every convergent germ.

36. Ring homomorphism

A homomorphism of commutative rings with identity is a map $f : R \rightarrow S$ such that $f(a+b) = f(a) + f(b)$, $f(ab) = f(a)f(b)$ for all $a, b \in R$, and $f(1_R) = 1_S$.

Prerequisites: Definition 33.

Example.

At $p = [1 : 0] \in \mathbf{CP}^1$, evaluation $\mathcal{O}_{X,p} \rightarrow \mathbb{C}$, $[f]_p \mapsto f(p)$, is a unital ring homomorphism: $[fg]_p$ maps to $f(p)g(p)$ and $[1]_p$ maps to 1.

37. Localization of a ring

Let R be a commutative ring with identity and $S \subseteq R$ a subset containing 1 and closed under multiplication. The localization $S^{-1}R$ consists of equivalence classes a/s of pairs $(a, s) \in R \times S$, where

$$(a, s) \sim (b, t) \iff \text{there is } u \in S \text{ such that } u(at - bs) = 0.$$

Addition and multiplication are $(a/s) + (b/t) = (at + bs)/(st)$ and $(a/s)(b/t) = ab/(st)$. The canonical homomorphism is $a \mapsto a/1$. An ideal $\mathfrak{p} \neq R$ is prime if $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$; put $R_{\mathfrak{p}} = (R \setminus \mathfrak{p})^{-1}R$. If $0 \in S$, the localization is the zero ring.

Prerequisites: Definition 35, Definition 36.

Example.

On the algebraic chart of $\mathbf{CP}^1$, take $R = \mathbb{C}[z]$ and $S = \{b : b(0) \neq 0\}$. Then $R_{(z)} = \{a(z)/b(z) : b(0) \neq 0\}$ is the algebraic local ring at 0; $1/(1-z)$ belongs to it but $1/z$ does not. It embeds in the analytic local ring $\mathbb{C}\{z\}$, and these rings must not be identified: e^z is an analytic germ but not a rational function.

38. Homomorphism of abelian groups

A homomorphism of abelian groups is a map $f : A \rightarrow B$ satisfying $f(a+a') = f(a) + f(a')$ for every $a, a' \in A$.

Prerequisites: Definition 32.

Example.

Evaluation at N gives an abelian-group homomorphism $\text{ev}_N : C^\infty(S^2, \mathbb{R}) \rightarrow \mathbb{R}$, $f \mapsto f(N)$. In particular $\text{ev}_N(2 + 3H) = 5$ and $\text{ev}_N(f + g) = f(N) + g(N)$.

39. Maximal ideal

A maximal ideal of a commutative ring R with identity is a proper ideal $\mathfrak{m} \subsetneq R$ such that every ideal J with $\mathfrak{m} \subseteq J \subseteq R$ equals either $\mathfrak{m}$ or R .

Prerequisites: Definition 35.

Example.

The ideal $(z) \subset \mathcal{O}_{\mathbf{CP}^1,0}$ is maximal because evaluation gives $\mathcal{O}_{\mathbf{CP}^1,0}/(z) \simeq \mathbb{C}$, a field. It consists exactly of germs vanishing at 0.

40. Local ring

A local ring is a commutative ring with identity having exactly one maximal ideal.

Prerequisites: Definition 39.

Example.

A germ $f \in \mathcal{O}_{\mathbf{CP}^1,0}$ is a unit exactly when $f(0) \neq 0$, since its reciprocal is then holomorphic near 0. Thus all nonunits form the single maximal ideal (z) , making this ring local.

41. Category

A category consists of objects, a set $\text{Hom}(A, B)$ for each ordered pair of objects, identity morphisms Id_A , and associative composition maps

$$\text{Hom}(B, C) \times \text{Hom}(A, B) \longrightarrow \text{Hom}(A, C), \quad (g, f) \longmapsto g \circ f,$$

such that $\text{Id}_B \circ f = f = f \circ \text{Id}_A$ whenever $f : A \rightarrow B$.

Example.

Fix $X = \mathbf{CP}^1$. In $\text{Sh}(X, \mathbf{Ab})$, objects include $\mathbb{Z}$, $\mathcal{O}_X$ under addition, and the skyscraper $i_*\mathbb{C}$ at $p = 0$. Morphisms are compatible families of homomorphisms on open sets. Evaluation $\mathcal{O}_X \rightarrow i_*\mathbb{C}$ is one such morphism.

42. Isomorphism

A morphism $f : A \rightarrow B$ in a category is an isomorphism if there is a morphism $g : B \rightarrow A$ with $g \circ f = \text{Id}_A$ and $f \circ g = \text{Id}_B$.

Prerequisites: Definition 41.

Example.

On $\mathbf{CP}^1$, the map $\mathcal{O}_X(-2[\infty]) \rightarrow \mathcal{K}_X$, $f \mapsto f dz$, is a sheaf isomorphism. On U_0 it sends 1 to dz ; on U_∞ it sends w^2 to $-dw$. Both images are local frames, giving local inverses which agree on overlaps.

43. Real and complex vector space

Fix $k = \mathbb{R}$ or $\mathbb{C}$. A k -vector space is an abelian group V with scalar multiplication $k \times V \rightarrow V$ such that $a(u + v) = au + av$, $(a + b)v = av + bv$, $(ab)v = a(bv)$, and $1v = v$, for all $a, b \in k$ and $u, v \in V$. A k -linear map preserves addition and multiplication by every scalar in k .

Prerequisites: Definition 32.

Example.

At $N = (0, 0, 1) \in S^2$, $T_N S^2 = \{(a, b, 0) : a, b \in \mathbb{R}\}$ is a real vector space with basis e_1, e_2 . Its complexification has basis e_1, e_2 over $\mathbb{C}$, while $T_N^{1,0} \mathbf{CP}^1$ has complex dimension one. The scalar field changes the dimension statement.

44. Dual, tensor product and exterior power

For k -vector spaces V, W , the dual is $V^* = \text{Hom}_k(V, k)$. A tensor product is a vector space $V \otimes_k W$ with a bilinear map $b : V \times W \rightarrow V \otimes_k W$ such that every bilinear map $f : V \times W \rightarrow Z$ to any k -vector space factors uniquely as $f = \tilde{f} \circ b$ with $\tilde{f}$ linear. For $q \geq 1$, the exterior power $\bigwedge^q V$ has an alternating q -linear map $V^q \rightarrow \bigwedge^q V$ with the analogous universal property for alternating q -linear maps; alternating means zero whenever two inputs coincide. Put $\bigwedge^0 V = k$.

Prerequisites: Definition 43.

Example.

At $N \in S^2$, take the tangent basis e_1, e_2 and dual basis ϵ^1, ϵ^2 . Then $(\epsilon^1 \otimes \epsilon^2)(e_1, e_2) = 1$, while $(\epsilon^1 \wedge \epsilon^2)(e_1, e_2) = 1$ and its value on (e_2, e_1) is -1 . The first tensor is not alternating, and the dual covector satisfies $\epsilon^1(e_1) = 1$.

45. Covariant functor

A covariant functor $T : \mathcal{C} \rightarrow \mathcal{D}$ assigns an object $T(A)$ to each object A and a morphism $T(f) : T(A) \rightarrow T(B)$ to each morphism $f : A \rightarrow B$, satisfying $T(\text{Id}_A) = \text{Id}_{T(A)}$ and $T(g \circ f) = T(g) \circ T(f)$ for all composable morphisms.

Prerequisites: Definition 41.

Example.

On $X = \mathbf{CP}^1$, $\Gamma(X, -)$ is covariant: a sheaf map $\phi : \mathcal{F} \rightarrow \mathcal{G}$ gives $\phi_X : \mathcal{F}(X) \rightarrow \mathcal{G}(X)$. The sheaf map $\times 2 : \underline{\mathbb{Z}} \rightarrow \underline{\mathbb{Z}}$ is sent to $\times 2 : \mathbb{Z} \rightarrow \mathbb{Z}$.

46. Finite direct sum / Biproduct

In a category whose morphism sets are abelian groups and whose composition is bilinear, a biproduct of A, B is an object $A \oplus B$ with maps i_A, i_B, p_A, p_B of the indicated inclusion and projection directions satisfying

$$p_A i_A = \text{Id}_A, \quad p_B i_B = \text{Id}_B, \quad p_A i_B = p_B i_A = 0, \quad i_A p_A + i_B p_B = \text{Id}_{A \oplus B}.$$

Finite direct sums are defined by iteration; the empty direct sum is a zero object.

46. Finite direct sum / Biproduct (continued)

Prerequisites: Definition 41, Definition 32.

Example.

On $\mathbf{CP}^1$, $\mathcal{O}_X \oplus \mathcal{K}_X$ has local sections (f, α) . The inclusions $f \mapsto (f, 0)$, $\alpha \mapsto (0, \alpha)$ and coordinate projections satisfy $p_i \iota_j = \delta_{ij}$ and $\iota_1 p_1 + \iota_2 p_2 = \text{Id}$.

47. Zero object

A zero object in a category is an object 0 such that for every object A there is exactly one morphism $0 \rightarrow A$ and exactly one morphism $A \rightarrow 0$. The composite $A \rightarrow 0 \rightarrow B$ is the zero morphism $A \rightarrow B$.

Prerequisites: Definition 41.

Example.

The zero sheaf on S^2 assigns the zero abelian group to every open set, including the empty set. There is exactly one sheaf map $0 \rightarrow \mathcal{F}$ and exactly one $\mathcal{F} \rightarrow 0$ for any abelian sheaf $\mathcal{F}$.

48. Categorical kernel

In a category with abelian morphism groups and bilinear composition, a kernel of $f : A \rightarrow B$ is a morphism $k : K \rightarrow A$ with $fk = 0$ such that for every object T and every morphism $g : T \rightarrow A$ with $fg = 0$, there is a unique $\tilde{g} : T \rightarrow K$ satisfying $g = k\tilde{g}$.

Prerequisites: Definition 47.

Example.

For $i : \{0\} \hookrightarrow \mathbf{CP}^1$, the categorical kernel of evaluation $\mathcal{O}_X \rightarrow i_*\mathbb{C}$ is the ideal sheaf $\mathcal{I}_0$: near 0 its sections are $zf(z)$, and away from 0 it equals $\mathcal{O}_X$. A sheaf map killed by evaluation factors uniquely through this inclusion.

49. Categorical cokernel

In a category with abelian morphism groups and bilinear composition, a cokernel of $f : A \rightarrow B$ is a morphism $q : B \rightarrow Q$ with $qf = 0$ such that for every object T and every morphism $g : B \rightarrow T$ with $gf = 0$, there is a unique $\tilde{g} : Q \rightarrow T$ satisfying $g = \tilde{g}q$.

Prerequisites: Definition 47.

Example.

The cokernel of $\mathcal{I}_0 \hookrightarrow \mathcal{O}_X$ on $\mathbf{CP}^1$ is $i_*\mathbb{C}$, with quotient map evaluation. A map from $\mathcal{O}_X$ vanishing on $\mathcal{I}_0$ factors uniquely through the value at 0 .

50. Categorical image

In an additive category with kernels and cokernels, the image of $f : A \rightarrow B$ is the kernel $\text{im } f = \ker(B \rightarrow \text{coker } f)$, together with its kernel morphism into B .

Prerequisites: Definition 48, Definition 49.

Example.

On S^2 , the image sheaf of $\times 2 : \underline{\mathbb{Z}} \rightarrow \underline{\mathbb{Z}}$ is $2\underline{\mathbb{Z}}$. Its sections are locally constant even integers. It is the kernel of the quotient to $\underline{\mathbb{Z}/2\mathbb{Z}}$.

51. Coimage

In an additive category with kernels and cokernels, the coimage of $f : A \rightarrow B$ is $\text{coim } f = \text{coker}(\ker f \rightarrow A)$, together with its quotient morphism from A .

Prerequisites: Definition 48, Definition 49.

Example.

The coimage of $\times 2 : \underline{\mathbb{Z}} \rightarrow \underline{\mathbb{Z}}$ on S^2 is $\underline{\mathbb{Z}}/0$. The canonical coimage-to-image map sends $n \mapsto 2n$ and is an isomorphism onto $2\underline{\mathbb{Z}}$, even though the original map is not onto $\underline{\mathbb{Z}}$.

52. Monomorphism

A morphism $j : A \rightarrow B$ is a monomorphism if, for every object T and morphisms $g, h : T \rightarrow A$, the equality $gj = jh$ implies $g = h$.

Prerequisites: Definition 41.

Example.

Multiplication by 2 on $\underline{\mathbb{Z}}$ over S^2 is a monomorphism: if $2a = 2b$ for locally constant integer sections then $a = b$. Its stalk at every point is the injective map $\mathbb{Z} \rightarrow \mathbb{Z}$, $n \mapsto 2n$.

53. Abelian category

An abelian category is a category in which every morphism set is an abelian group, composition is bilinear, a zero object and all finite biproducts exist, and every morphism has a kernel and a cokernel, with the following additional property. For every $f : A \rightarrow B$, the canonical factorization

$$A \xrightarrow{p_f} \text{coim } f \xrightarrow{\bar{f}} \text{im } f \xrightarrow{i_f} B, \quad f = i_f \bar{f} p_f,$$

has $\bar{f}$ an isomorphism. The arrows p_f, i_f are the cokernel of $\ker f \rightarrow A$ and the kernel of $B \rightarrow \text{coker } f$, respectively.

Prerequisites: Definition 46, Definition 48, Definition 49, Definition 50, Definition 51, Definition 52.

Example.

53. Abelian category (continued)

The category of abelian sheaves on S^2 is abelian. The concrete morphism $\times 2 : \underline{\mathbb{Z}} \rightarrow \underline{\mathbb{Z}}$ has kernel 0, image $2\underline{\mathbb{Z}}$, and cokernel $\underline{\mathbb{Z}}/2\underline{\mathbb{Z}}$, and its coimage maps isomorphically onto its image. These illustrate the axioms; the assertion for every morphism is the general abelian-sheaf theorem.

54. Kernel of a homomorphism

For a homomorphism $f : A \rightarrow B$ of abelian groups, $\ker f = \{a \in A : f(a) = 0_B\}$ is a subgroup of A .

Prerequisites: Definition 38.

Example.

For ev_N on S^2 , $\ker \text{ev}_N = \{f : f(N) = 0\}$. The nonzero function $H - 1$ belongs to this kernel, since $H(N) = 1$.

55. Image of a homomorphism

For a homomorphism $f : A \rightarrow B$ of abelian groups, $\text{im } f = \{f(a) : a \in A\}$ is a subgroup of B .

Prerequisites: Definition 38.

Example.

The image of $\text{ev}_N : C^\infty(S^2, \mathbb{R}) \rightarrow \mathbb{R}$ is all of $\mathbb{R}$: every $a \in \mathbb{R}$ is the image of the constant function $f \equiv a$.

56. Quotient group

If K is a subgroup of an abelian group A , the quotient group is $A/K = \{a + K : a \in A\}$, where $a + K = b + K$ exactly when $a - b \in K$, with addition $(a + K) + (b + K) = (a + b) + K$.

Prerequisites: Definition 32.

Example.

The quotient $C^\infty(S^2, \mathbb{R})/\ker \text{ev}_N$ is isomorphic to $\mathbb{R}$ by $[f] \mapsto f(N)$. Thus H and the constant function 1 define the same quotient class although they are different functions on S^2 .

57. Hermitian inner product

A Hermitian inner product on a complex vector space V is a function $h : V \times V \rightarrow \mathbb{C}$ which is complex-linear in its first argument, conjugate-linear in its second, and satisfies $h(u, v) = \overline{h(v, u)}$ for all $u, v \in V$ and $h(v, v) > 0$ for all $v \neq 0$.

Prerequisites: Definition 43.

Example.

57. Hermitian inner product (continued)

In the complex tangent line $T_0^{1,0}\mathbf{CP}^1 = \mathbb{C}\partial_z$, set $h(a\partial_z, b\partial_z) = 2a\bar{b}$. Thus $h(\partial_z, i\partial_z) = -2i$, $h(i\partial_z, \partial_z) = 2i$, and $h(a\partial_z, a\partial_z) = 2|a|^2 > 0$ for $a \neq 0$.

58. Cokernel of a homomorphism

The cokernel of a homomorphism $f : A \rightarrow B$ of abelian groups is $\text{coker } f = B/\text{im } f$, with the quotient homomorphism $B \rightarrow B/\text{im } f$.

Prerequisites: Definition 55, Definition 56.

Example.

On the connected sphere, $\Gamma(S^2, \mathbb{Z}) = \mathbb{Z}$. The map induced by multiplication by 2 has cokernel $\mathbb{Z}/2\mathbb{Z}$; the class of the constant section 1 is nonzero and twice that class is zero.

59. Contravariant functor

A contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ is a functor $T : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$: it assigns $T(f) : T(B) \rightarrow T(A)$ to $f : A \rightarrow B$, preserves identities, and satisfies $T(gf) = T(f)T(g)$. The opposite category $\mathcal{C}^{\text{op}}$ has the same objects and all arrows reversed.

Prerequisites: Definition 41.

Example.

On $X = \mathbf{CP}^1$, the functor $\text{Hom}(-, \mathcal{O}_X)$ on $\mathcal{O}_X$ -modules is contravariant. Precomposition with $\times 2 : \mathcal{O}_X \rightarrow \mathcal{O}_X$ sends a map ψ to $\psi \circ (\times 2)$; the arrow between Hom spaces goes in the reversed direction.

Differential and complex geometry

60. Holomorphic function

A function $f : U \rightarrow \mathbb{C}$ on an open subset $U \subseteq \mathbb{C}$ is holomorphic if the limit $\lim_{h \rightarrow 0} (f(z+h) - f(z))/h$ exists in $\mathbb{C}$ at every $z \in U$.

Example.

On $U_0 \simeq \mathbb{C} \subset \mathbf{CP}^1$, $f(z) = z^2$ is holomorphic, with derivative $2z$. The function $\bar{z}$ is not holomorphic. A holomorphic function on all of the compact connected $\mathbf{CP}^1$ must be constant; a chart function need not extend globally.

61. Riemann surface

A Riemann surface is a complex manifold of complex dimension one: a Hausdorff second-countable space with a maximal atlas of homeomorphisms to open subsets of $\mathbb{C}$, having holomorphic transition maps. Connectedness, when required, is an additional hypothesis.

Prerequisites: Definition 25, Definition 26, Definition 27, Definition 60.

61. Riemann surface (continued)

Example.

The sphere with coordinates $z = (X + iY)/(1 + Z)$ and $w = (X - iY)/(1 - Z)$ is a Riemann surface. The overlap relation $w = 1/z$ is holomorphic with nonzero derivative for $z \neq 0$. This complex structure identifies it with $\mathbf{CP}^1$.

62. Holomorphic tangent and cotangent bundles

On a complex n -manifold X , the holomorphic tangent bundle $T^{1,0}X$ has local frames $\partial/\partial z^1, \dots, \partial/\partial z^n$ and transition maps given by the complex Jacobian and the chain rule. It is a rank- n holomorphic vector bundle. Its fibrewise complex-linear dual is the holomorphic cotangent bundle $(T^{1,0}X)^*$, with dual frames $dz^1, \dots, dz^n$. A rank- r holomorphic vector bundle has fibrewise complex-linear biholomorphic trivializations over an open cover with fibre $\mathbb{C}^r$.

Prerequisites: Definition 9, Definition 8.3, Definition 44.

Example.

On $\mathbf{CP}^1$, $\partial_w = -z^2 \partial_z$ and $dw = -z^{-2} dz$. Pairing gives $dw(\partial_w) = 1$ on the overlap. Thus the tangent and cotangent transition functions are reciprocal, and each bundle has complex rank one.

63. Complex projective space

For $n \geq 0$, complex projective space is

$$\mathbf{CP}^n = (\mathbb{C}^{n+1} \setminus \{0\}) / \sim, \quad Z \sim W \iff W = \lambda Z \text{ for some } \lambda \in \mathbb{C}^\times.$$

Write its elements $[Z_0 : \dots : Z_n]$. On $U_i = \{Z_i \neq 0\}$, the coordinates $(Z_0/Z_i, \dots, \widehat{Z_i/Z_i}, \dots, Z_n/Z_i)$ give the standard complex-manifold structure.

Prerequisites: Definition 9, Definition 43.

Example.

Every point of $\mathbf{CP}^1$ is a complex line $\mathbb{C}(S, T) \subset \mathbb{C}^2$, with $(S, T) \neq (0, 0)$. For example $[1 : 2] = [3 : 6]$, while $[1 : 2] \neq [1 : 3]$. The point $[0 : 1]$ is omitted from the z -chart and is $w=0$ in the second chart.

64. Cotangent space

The cotangent space of a smooth manifold M at p is the real-linear dual $T_p^*M = \text{Hom}_{\mathbb{R}}(T_p M, \mathbb{R})$.

Prerequisites: Definition 8.3, Definition 44.

Example.

At $N \in S^2$, the covectors $\epsilon^1(a, b, 0) = a$ and $\epsilon^2(a, b, 0) = b$ form a basis of $T_N^*S^2$. Under our

64. Cotangent space (continued)

z -chart, $dx_N = \epsilon^1/2$ and $dy_N = \epsilon^2/2$, because $d\Phi_0(\partial_x) = 2e_1$.

65. Coordinate one-form

For a smooth chart $\varphi = (x^1, \dots, x^n) : U \rightarrow V$ and $v = [\gamma] \in T_p M$ with $p \in U$, define $dx_p^i(v) = (x^i \circ \gamma)'(0)$. This is independent of the representative curve and reads the i th coordinate of $(\varphi \circ \gamma)'(0)$. The family $(dx_p^i)_{p \in U}$ is the coordinate one-form dx^i .

Prerequisites: Definition 18.

Example.

In the chart $z = x + iy$ of $\mathbf{CP}^1$, the real coordinate forms satisfy $dx(\partial_x) = 1$, $dx(\partial_y) = 0$, $dy(\partial_x) = 0$, $dy(\partial_y) = 1$. The complex coordinate form is $dz = dx + i dy$, and $dw = -z^{-2}dz$ on the overlap.

66. Wedge product / Exterior product

For $\alpha \in \Omega^p(M)$ and $\beta \in \Omega^q(M)$ with $p, q \geq 0$, their wedge product is defined, at $x \in M$, by

$$\begin{aligned} & (\alpha \wedge \beta)_x(v_1, \dots, v_{p+q}) \\ &= \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} \text{sgn}(\sigma) \alpha_x(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \beta_x(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)}). \end{aligned}$$

Here S_{p+q} is the permutation group, and a 0-form contributes its value at x . In particular $(dx^{i_1} \wedge \dots \wedge dx^{i_q})(v_1, \dots, v_q) = \det(dx^{i_a}(v_b))_{a,b}$.

Prerequisites: Definition 18.

Example.

On the z -chart of $\mathbf{CP}^1$, $dz \wedge d\bar{z} = (dx + i dy) \wedge (dx - i dy) = -2i dx \wedge dy$. In particular $dz \wedge dz = 0$, whereas the tensor $dz \otimes dz$ is nonzero. Wedge and tensor products are different operations.

67. Primitive of a differential form

For a smooth q -form α on M with $q \geq 1$, a primitive is a smooth $(q-1)$ -form β on M satisfying $d\beta = \alpha$.

Prerequisites: Definition 19.

Example.

For the form $\alpha = dH$ on S^2 , both H and $H + 7$ are global primitives. On the connected sphere any two function primitives differ by a constant. On $U_0 \cap U_\infty \simeq \mathbb{C}^\times$, dz/z has local logarithmic primitives but no global single-valued primitive.

Sheaves and local data

68. Presheaf of abelian groups

A presheaf of abelian groups on a topological space X assigns an abelian group $\mathcal{F}(U)$ to each open $U \subseteq X$ and a homomorphism $\rho_{UV} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$ to each open inclusion $V \subseteq U$, with $\rho_{UU} = \text{Id}$ and $\rho_{UV} = \rho_{VW}\rho_{UV}$ whenever $W \subseteq V \subseteq U$. Write $s|_V = \rho_{UV}(s)$.

Prerequisites: Definition 23, Definition 38.

Example.

On $X = \mathbf{CP}^1$, assign to every open U the additive group $\mathcal{O}_X(U)$ of holomorphic functions. For $V \subset U$, use ordinary restriction. Restricting z^2 first from U_0 to $|z| < 2$ and then to $|z| < 1$ equals direct restriction. This is an abelian presheaf (and in fact a sheaf).

69. Sheaf

A presheaf $\mathcal{F}$ on X is a sheaf if for every open $U \subseteq X$, every open cover $U = \bigcup_i U_i$, and every family $s_i \in \mathcal{F}(U_i)$ with $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j , there is a unique $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i . The empty cover is included, so an abelian sheaf has $\mathcal{F}(\emptyset) = 0$.

Prerequisites: Definition 68.

Example.

On the two charts of $\mathbf{CP}^1$, take $f_0 = 3$ and $f_\infty = 3$. Their restrictions agree, so the sheaf axiom glues them uniquely to the global constant function 3. Choosing $f_0 = z$ and $f_\infty = 0$ does not meet the overlap condition, so the axiom does not assert that these functions glue.

70. Sheaf of rings

A sheaf of commutative rings with identity on X assigns such a ring $\mathcal{O}(U)$ to each open U , with identity-preserving ring homomorphisms as restriction maps, satisfying the presheaf identities and the existence and uniqueness of gluing for every open cover.

Prerequisites: Definition 69, Definition 36.

Example.

The holomorphic function sheaf $\mathcal{O}_X$ on $\mathbf{CP}^1$ is a sheaf of unital commutative rings: restriction preserves addition, multiplication, and the constant function 1. For example $(fg)|_V = (f|_V)(g|_V)$ for $V \subset U_0$.

71. Section

For a presheaf $\mathcal{F}$ on X and an open subset $U \subseteq X$, a section over U is an element $s \in \mathcal{F}(U)$. Its restriction to an open subset $V \subseteq U$ is $s|_V = \rho_{UV}(s) \in \mathcal{F}(V)$.

Prerequisites: Definition 68.

Example.

71. Section (continued)

The function $s(z) = e^z$ belongs to $\mathcal{O}_X(U_0)$ on $\mathbf{CP}^1$: it is a section over the chart U_0 . No assertion that it is a section over all of X is being made; it has an essential singularity at infinity.

72. Germ

Fix $x \in X$ and a presheaf $\mathcal{F}$ on X . Consider pairs (U, s) with U an open neighbourhood of x and $s \in \mathcal{F}(U)$. Put $(U, s) \sim_x (V, t)$ if there is an open W with $x \in W \subseteq U \cap V$ and $s|_W = t|_W$. The equivalence class $[U, s]_x$, also written s_x , is the germ of s at x .

Prerequisites: Definition 71, Definition 24.

Example.

At $0 \in U_0 \subset \mathbf{CP}^1$, the function $1/(1 - z)$ on $|z| < 1/2$ and the sum $\sum_{n \geq 0} z^n$ on $|z| < 1$ define the same holomorphic germ. They agree on a smaller neighbourhood. The germ of 1 is different despite having the same value at 0.

73. Stalk

For a presheaf of abelian groups $\mathcal{F}$ and $x \in X$, the stalk $\mathcal{F}_x$ is the set of germs $[U, s]_x$ over all open neighbourhoods $U \ni x$. Two pairs are equivalent when their restrictions agree on some common open neighbourhood of x . Its abelian group structure is

$$[U, s]_x + [V, t]_x = [U \cap V, s|_{U \cap V} + t|_{U \cap V}]_x, \quad -[U, s]_x = [U, -s]_x,$$

with zero $[X, 0]_x$. Equivalently, $\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U)$.

Prerequisites: Definition 72.

Example.

At $0 \in \mathbf{CP}^1$, the stalk $\mathcal{O}_{X,0}$ is $\mathbb{C}\{z\}$, the convergent power-series ring. The elements 1 and $1 + z$ are distinct in this stalk; they become equal after passing to its residue field $\mathcal{O}_{X,0}/(z)$.

74. Ringed space

A ringed space is a pair $(X, \mathcal{O}_X)$ consisting of a topological space and a sheaf of commutative rings with identity, called its structure sheaf. It is locally ringed if every stalk $\mathcal{O}_{X,x}$ is a local ring.

Prerequisites: Definition 70, Definition 73, Definition 40.

Example.

The analytic $\mathbf{CP}^1$ with its sheaf $\mathcal{O}_X$ is a locally ringed space. At a finite point a the stalk has maximal ideal $(z - a)$; at infinity it has maximal ideal (w) . The underlying topological sphere alone does not specify this holomorphic structure sheaf.

75. Holomorphic structure sheaf

For a complex manifold X , its holomorphic structure sheaf is

$$\mathcal{O}_X(U) = \{f : U \rightarrow \mathbb{C} : f \text{ is holomorphic in every holomorphic chart}\}$$

for each open $U \subseteq X$, with pointwise addition and multiplication and ordinary restriction of functions.

Prerequisites: Definition 9, Definition 69, Definition 70.

Example.

On the z -chart of $\mathbf{CP}^1$, z^2 is a section of the holomorphic structure sheaf but $\bar{z}$ is not. Both are sections of the smooth complex function sheaf. This tests which structure sheaf is being used.

76. Morphism of sheaves

For sheaves of abelian groups $\mathcal{F}, \mathcal{G}$ on the same space X , a morphism $f : \mathcal{F} \rightarrow \mathcal{G}$ is a family of homomorphisms $f_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$, one for each open U , such that

$$\rho_{UV}^{\mathcal{G}} f_U = f_V \rho_{UV}^{\mathcal{F}} \quad (V \subseteq U \text{ open}).$$

Prerequisites: Definition 69, Definition 38.

Example.

On $\mathbf{CP}^1$, differentiation gives a sheaf map of complex vector spaces $d : \mathcal{O}_X \rightarrow \mathcal{K}_X$, with $f \mapsto df$. It commutes with restriction. It is not $\mathcal{O}_X$ -linear, since $d(fg) = f dg + g df$.

77. Sheaf of modules

On a ringed space $(X, \mathcal{O}_X)$, a sheaf of $\mathcal{O}_X$ -modules is a sheaf of abelian groups $\mathcal{E}$ with an $\mathcal{O}_X(U)$ -module structure on each $\mathcal{E}(U)$, satisfying $(as)|_V = (a|_V)(s|_V)$ for all open $V \subseteq U$, $a \in \mathcal{O}_X(U)$, and $s \in \mathcal{E}(U)$. A morphism of module sheaves is a sheaf morphism whose component over every U is $\mathcal{O}_X(U)$ -linear.

Prerequisites: Definition 34, Definition 74, Definition 76.

Example.

On $\mathbf{CP}^1$, $\mathcal{K}_X$ is an $\mathcal{O}_X$ -module sheaf: locally $f \cdot (a dz) = (fa) dz$, and restricting a product gives the product of the restrictions. For example $z \cdot ((1+z)dz) = (z+z^2)dz$.

78. Restriction of a sheaf

If U is open in X and $\mathcal{F}$ is a sheaf on X , its restriction $\mathcal{F}|_U$ is the sheaf on U defined by $(\mathcal{F}|_U)(V) = \mathcal{F}(V)$ for open $V \subseteq U$, with the original restriction maps.

Prerequisites: Definition 68.

78. Restriction of a sheaf (continued)

Example.

Restrict the holomorphic function sheaf of $\mathbf{CP}^1$ to U_0 . For every open $V \subset U_0$, $(\mathcal{O}_X|_{U_0})(V) = \mathcal{O}_X(V)$; the coordinate z identifies this restricted sheaf with the usual holomorphic function sheaf on $\mathbb{C}$.

79. Locally free sheaf

For $r \geq 0$, a sheaf $\mathcal{E}$ of $\mathcal{O}_X$ -modules is locally free of rank r if every $x \in X$ has an open neighbourhood U and an isomorphism of $\mathcal{O}_X|_U$ -module sheaves

$$\mathcal{E}|_U \simeq (\mathcal{O}_X|_U)^{\oplus r}.$$

This is an isomorphism on all open subsets of U , compatible with restriction, rather than just a fibrewise or stalkwise choice of bases.

Prerequisites: Definition 77, Definition 78, Definition 42.

Example.

On $\mathbf{CP}^1$, $\mathcal{K}_X|_{U_0} = \mathcal{O}_{U_0} dz$ and $\mathcal{K}_X|_{U_\infty} = \mathcal{O}_{U_\infty} dw$. It is locally free of rank one, but not globally free: a global basis would be a nowhere-zero holomorphic one-form, and no nonzero global holomorphic one-form exists on $\mathbf{CP}^1$.

80. Invertible sheaf

On a locally ringed space $(X, \mathcal{O}_X)$, an invertible sheaf is a locally free $\mathcal{O}_X$ -module sheaf of rank one. Thus every point has an open neighbourhood U with $\mathcal{L}|_U \simeq \mathcal{O}_X|_U$. A local generator e gives, for every open $V \subseteq U$, a unique expression $ae|_V$ for each section of $\mathcal{L}(V)$, with $a \in \mathcal{O}_X(V)$.

Prerequisites: Definition 79, Definition 44.

Example.

On $\mathbf{CP}^1$, evaluation gives $\mathcal{K}_X \otimes_{\mathcal{O}_X} \mathcal{T}_X \simeq \mathcal{O}_X$, where $\mathcal{T}_X$ is the sheaf of holomorphic tangent vectors. Locally $dz \otimes \partial_z \mapsto 1$; the reciprocal transition functions make these maps agree. Thus $\mathcal{K}_X$ is invertible.

81. Global section

For a sheaf $\mathcal{F}$ on X , its group of global sections is $\Gamma(X, \mathcal{F}) = \mathcal{F}(X)$.

Prerequisites: Definition 71.

Example.

For $X = \mathbf{CP}^1$, $\Gamma(X, \mathcal{O}_X) = \mathbb{C}$, represented by constant holomorphic functions. In contrast $\Gamma(X, \mathcal{K}_X) = 0$. Locally dz is a frame, but it is not a global holomorphic one-form.

82. Induced map on stalks

Let $\mathcal{F}, \mathcal{G}$ be sheaves of abelian groups on X , let $f : \mathcal{F} \rightarrow \mathcal{G}$ be a sheaf morphism, and fix $x \in X$. For each open U , write $f_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ for its component. The stalk $\mathcal{F}_x$ consists of equivalence classes $[U, s]_x$ of pairs with $x \in U$ and $s \in \mathcal{F}(U)$; pairs $(U, s), (V, t)$ are equivalent when $s|_W = t|_W$ for some open $x \in W \subseteq U \cap V$. Define $\mathcal{G}_x$ similarly. The induced homomorphism on stalks is

$$f_x : \mathcal{F}_x \longrightarrow \mathcal{G}_x, \quad [U, s]_x \longmapsto [U, f_U(s)]_x.$$

The identity $f_W(s|_W) = f_U(s)|_W$ ensures that equivalent representatives have equivalent images.

Prerequisites: Definition 76, Definition 73.

Example.

At $0 \in \mathbf{CP}^1$, the preceding map induces $d_0 : \mathcal{O}_{X,0} \rightarrow \mathcal{K}_{X,0}$, $[f]_0 \mapsto [df]_0$. Concretely $[z^3 + 2z]_0$ maps to $[(3z^2 + 2)dz]_0$. Agreement on a neighbourhood implies agreement of derivatives on that neighbourhood, so representatives do not change the output.

83. Exact sequence of sheaves

A sequence of sheaves of abelian groups on X ,

$$\dots \longrightarrow \mathcal{F}^{r-1} \xrightarrow{a^{r-1}} \mathcal{F}^r \xrightarrow{a^r} \mathcal{F}^{r+1} \longrightarrow \dots,$$

is exact at $\mathcal{F}^r$ if $\text{im } a_x^{r-1} = \ker a_x^r$ in $\mathcal{F}_x^r$ for every $x \in X$. It is exact if it is exact at every term. The maps a_x^r are the induced maps on stalks; exactness of sections over every open set is not required.

Prerequisites: Definition 82, Definition 118.

Example.

On the analytic $\mathbf{CP}^1$, $0 \rightarrow \mathbb{C} \rightarrow \mathcal{O}_X \xrightarrow{d} \mathcal{K}_X \rightarrow 0$ is exact as a sequence of complex-vector-space sheaves. The kernel consists of locally constant functions. On a small coordinate disc every holomorphic $a(z)dz$ has a holomorphic primitive, proving surjectivity on stalks.

84. Monomorphism of sheaves

A morphism $f : \mathcal{F} \rightarrow \mathcal{G}$ of sheaves of abelian groups on X is a monomorphism if for every $x \in X$ the induced map $f_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is injective: $f_x(a) = f_x(b)$ implies $a = b$ for all $a, b \in \mathcal{F}_x$.

Prerequisites: Definition 82.

Example.

The inclusion $\mathbb{C} \hookrightarrow \mathcal{O}_X$ on $\mathbf{CP}^1$ is injective on every stalk: a locally constant germ that becomes the zero holomorphic germ is already zero. This supplies a concrete sheaf monomorphism.

85. Epimorphism of sheaves

A morphism $f : \mathcal{F} \rightarrow \mathcal{G}$ of sheaves of abelian groups on X is an epimorphism if for every $x \in X$ the induced map $f_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is surjective: every $b \in \mathcal{G}_x$ equals $f_x(a)$ for some $a \in \mathcal{F}_x$. Surjectivity of f_U on every open set U is not required.

Prerequisites: Definition 82.

Example.

The derivative $d : \mathcal{O}_X \rightarrow \mathcal{K}_X$ on $\mathbf{CP}^1$ is surjective on stalks. It need not be surjective on sections of every open set: on $U_0 \cap U_\infty = \mathbb{C}^\times$, the form dz/z has no global holomorphic primitive because its integral around the unit circle is $2\pi i$.

86. Locally constant function

A function $f : U \rightarrow B$ from a topological space to a set is locally constant if every $x \in U$ has an open neighbourhood $V \subseteq U$ on which f is constant.

Prerequisites: Definition 24.

Example.

On the disconnected open subset $U = \{Z > 1/2\} \cup \{Z < -1/2\}$ of S^2 , define $f=1$ on the north cap and $f=2$ on the south cap. It is locally constant, but not constant on U . On the connected whole sphere a locally constant function is constant.

87. Constant sheaf

For an abelian group B , the constant sheaf $\underline{B}$ on X assigns to an open U the abelian group of locally constant functions $U \rightarrow B$, with pointwise addition and restriction of functions. These functions need not be constant on all of U .

Prerequisites: Definition 86, Definition 69.

Example.

The constant sheaf $\underline{\mathbb{Z}}$ on S^2 assigns locally constant integer-valued functions to U . On the two-cap open set from the preceding example its sections form $\mathbb{Z} \oplus \mathbb{Z}$, whereas on S^2 its sections form $\mathbb{Z}$. This is not the presheaf assigning a single copy of $\mathbb{Z}$ to every nonempty open set.

88. Sheaf of differential forms

For a smooth manifold M and an integer $q \geq 0$, the sheaf of smooth real differential q -forms is $\mathcal{A}^q(U) = \Omega^q(U)$ for open $U \subseteq M$, with restriction of forms as restriction maps.

Prerequisites: Definition 18, Definition 69.

Example.

The sheaf $\mathcal{A}^2$ on S^2 assigns smooth two-forms to each open U . Its global section $\Omega_p(u, v) = p \cdot (u \times v)$ restricts in the z -chart to $4(1 + |z|^2)^{-2} dx \wedge dy$. Its local coordinate formulas agree

88. Sheaf of differential forms (continued)

by pullback on overlaps.

89. Sheaf of holomorphic functions

For a Riemann surface X , the sheaf of holomorphic functions assigns to each open $U \subseteq X$ the ring $\mathcal{O}_X(U)$ of complex-valued functions holomorphic in local complex coordinates, with pointwise operations and restriction of functions.

Prerequisites: Definition 61, Definition 70.

Example.

On $\mathbf{CP}^1$, $\mathcal{O}_X(U_0) = \mathcal{O}(\mathbb{C})$ contains e^z , but $\mathcal{O}_X(X) = \mathbb{C}$. This distinguishes the sheaf, its sections on a chart, and its global sections in one explicit example.

90. Stalk of the holomorphic function sheaf

For x on a Riemann surface X , $\mathcal{O}_{X,x}$ is the ring of germs of holomorphic functions near x . Two representatives give the same germ if they agree on some common open neighbourhood of x . Addition and multiplication are performed after restricting representatives to a common neighbourhood.

Prerequisites: Definition 89, Definition 73.

Example.

Using z at $0 \in \mathbf{CP}^1$, the stalk of the holomorphic function sheaf consists of convergent germs $\sum_{n \geq 0} a_n z^n$. The representative $1/(1-z)$ corresponds to the germ $1 + z + z^2 + \cdots$, with positive radius of convergence.

91. Maximal ideal of the holomorphic local ring

For a Riemann surface X and $x \in X$, the maximal ideal of $\mathcal{O}_{X,x}$ is

$$\mathfrak{m}_x = \{[U, f]_x : x \in U, f \in \mathcal{O}_X(U), f(x) = 0\}.$$

The value $f(x)$ is independent of the representative of the germ.

Prerequisites: Definition 90, Definition 39.

Example.

At $0 \in \mathbf{CP}^1$, the maximal ideal of holomorphic germs is $\mathfrak{m}_0 = \{f : f(0) = 0\} = (z)$. The residue map takes $a_0 + a_1 z + \cdots$ to a_0 , so the residue field is $\mathbb{C}$.

92. Ring of convergent power series

The ring $\mathbb{C}\{t\}$ consists of power series $\sum_{n \geq 0} a_n t^n$ with $a_n \in \mathbb{C}$ for which some $r > 0$ exists such that $\sum a_n z^n$ converges for every $|z| < r$. Equality and addition are coefficientwise; the

92. Ring of convergent power series (continued)

coefficient of t^n in a product is $\sum_{i=0}^n a_i b_{n-i}$. Its identity is the constant series 1.

Prerequisites: Definition 60, Definition 33.

Example.

The analytic local ring at $0 \in \mathbf{CP}^1$ is $\mathbb{C}\{z\}$. The series $\sum_{n \geq 0} z^n$ represents a germ, but $\sum_{n \geq 0} n!z^n$ does not: its radius of convergence is zero.

93. Formal power series ring

The formal power series ring $\mathbb{C}[[t]]$ consists of all sequences $(a_n)_{n \geq 0}$ of complex numbers, written $\sum_{n \geq 0} a_n t^n$. Equality and addition are coefficientwise; multiplication is given by $(ab)_n = \sum_{i=0}^n a_i b_{n-i}$. The identity is the constant series 1. No convergence condition is imposed.

Prerequisites: Definition 33.

Example.

The formal series $\sum_{n \geq 0} n!z^n$ is an element of $\mathbb{C}[[z]]$, the completed formal coordinate ring at $0 \in \mathbf{CP}^1$. Its existence as a formal element makes no assertion of a holomorphic function near 0.

94. Power of an ideal

For an ideal J in a commutative ring R and $m \geq 1$,

$$J^m = \left\{ \sum_{k=1}^N a_{k,1} \cdots a_{k,m} : N \geq 0, a_{k,j} \in J \right\},$$

where the empty sum is zero. Thus J^m is generated by all products of m elements of J , not merely their individual m th powers.

Prerequisites: Definition 35.

Example.

At $0 \in \mathbf{CP}^1$, $\mathfrak{m}_0^k = (z^k) \subset \mathbb{C}\{z\}$. The germ $z^3 + z^4$ lies in $\mathfrak{m}_0^3$ but not $\mathfrak{m}_0^4$. This records a vanishing order of exactly three.

95. Jet

For a Riemann surface X , a point x , and $m \geq 1$, a jet of order $m-1$ of a holomorphic function at x is an element of $\mathcal{O}_{X,x}/\mathfrak{m}_x^m$. The jet of a germ g is $g + \mathfrak{m}_x^m$; two germs g, h have the same jet exactly when $g - h \in \mathfrak{m}_x^m$.

Prerequisites: Definition 94, Definition 90.

Example.

95. Jet (continued)

The second jet of e^z at $0 \in \mathbf{CP}^1$ is its class in $\mathcal{O}_{X,0}/\mathfrak{m}_0^3$: $[e^z] = [1 + z + z^2/2]$. Terms of degree at least three are forgotten, while value, first derivative and second derivative remain.

96. Completion

Let $R = \mathcal{O}_{X,x}$ for a point x of a Riemann surface, with maximal ideal $\mathfrak{m}_x$. Put $R_m = R/\mathfrak{m}_x^m$, and let $\pi_m : R_{m+1} \rightarrow R_m$ send $g + \mathfrak{m}_x^{m+1}$ to $g + \mathfrak{m}_x^m$. The completion is the ring

$$\hat{R} = \varprojlim_m R_m = \{(j_m) \in \prod_{m \geq 1} R_m : \pi_m(j_{m+1}) = j_m \text{ for every } m \geq 1\},$$

with componentwise addition and multiplication.

Prerequisites: Definition 94.

Example.

At $0 \in \mathbf{CP}^1$, the compatible truncations $1 + z + \cdots + z^k$ modulo z^{k+1} determine the element $\sum_{n \geq 0} z^n$ in $\hat{\mathcal{O}}_{X,0} \simeq \mathbb{C}[[z]]$. The truncations of $\sum n!z^n$ also give an element of this completion even though no convergent germ has those coefficients.

Line bundles and divisors

97. Holomorphic line bundle and local trivialization

A holomorphic line bundle on a complex manifold X consists of a complex manifold L , a holomorphic map $\pi : L \rightarrow X$, a one-dimensional complex vector-space structure on every fibre $L_x = \pi^{-1}(x)$, and local trivializations: an open cover (U_i) and biholomorphisms $\tau_i : \pi^{-1}(U_i) \rightarrow U_i \times \mathbb{C}$ such that $\text{pr}_1 \tau_i = \pi$ and each restriction to a fibre is complex-linear. The trivial line bundle is $X \times \mathbb{C} \rightarrow X$.

Prerequisites: Definition 9, Definition 43.

Example.

The canonical bundle on $\mathbf{CP}^1$ is trivialized over U_0 by $(p, a dz_p) \mapsto (p, a)$ and over U_∞ by $(p, b dw_p) \mapsto (p, b)$. Both trivializations are holomorphic and linear on fibers; their change is induced by $dw = -z^{-2}dz$.

98. Section and local frame

For a holomorphic line bundle $\pi : L \rightarrow X$, a holomorphic section over an open U is a holomorphic map $s : U \rightarrow L$ satisfying $\pi \circ s = \text{Id}_U$. A local frame is a section e with $e(x) \neq 0$ for all $x \in U$. Every holomorphic section on U then has a unique expression $s = fe$ with $f \in \mathcal{O}_X(U)$. Sections with pointwise operations and restriction form an $\mathcal{O}_X$ -module sheaf.

Prerequisites: Definition 97, Definition 75.

Example.

98. Section and local frame (continued)

On $U_0 \subset \mathbf{CP}^1$, $e = dz$ is a local frame of K_X , and $s = (z^2 + 1)dz$ is a holomorphic section. The section s is not a frame on all of U_0 , because it vanishes at $z = i$ and $z = -i$.

99. Transition function

For local holomorphic frames e_i of a line bundle over an open cover (U_i) , the transition functions $g_{ij} \in \mathcal{O}_X^*(U_i \cap U_j)$ are defined by $e_j = g_{ij}e_i$. Here $\mathcal{O}_X^*(U)$ is the multiplicative group of nowhere-zero holomorphic functions. They satisfy $g_{ii} = 1$ and $g_{ik} = g_{ij}g_{jk}$ on triple overlaps. If $s = f_i e_i = f_j e_j$, then $f_i = g_{ij}f_j$.

Prerequisites: Definition 98.

Example.

For $K_{\mathbf{CP}^1}$ take $e_0 = dz$, $e_\infty = dw$. The frame transition is $g_{0\infty} = -z^{-2}$. If $s = a(z)dz = b(w)dw$, the coefficient relation is $a(z) = -z^{-2}b(1/z)$, or $b(w) = -w^{-2}a(1/w)$.

100. Isomorphism of line bundles

An isomorphism of holomorphic line bundles $\pi_L : L \rightarrow X$ and $\pi_E : E \rightarrow X$ is a biholomorphism $\Phi : L \rightarrow E$ such that $\pi_E \Phi = \pi_L$ and $\Phi : L_x \rightarrow E_x$ is complex-linear for every x . The bundle L is trivial if it is isomorphic in this sense to $X \times \mathbb{C}$.

Prerequisites: Definition 97.

Example.

The bundle associated with $\mathcal{O}_X(-2[\infty])$ is isomorphic to $K_{\mathbf{CP}^1}$ by multiplication by the meromorphic section dz . Its local frames $1, w^2$ map to $dz, -dw$, respectively. Every fiber map is complex-linear and invertible, and the maps cover the identity of $\mathbf{CP}^1$.

101. Dual line bundle

The dual of a holomorphic line bundle L is the line bundle L^* with fibre $L_x^* = \text{Hom}_{\mathbb{C}}(L_x, \mathbb{C})$. If e_i are local frames, the dual frames satisfy $e_i^*(e_i) = 1$; under $e_j = g_{ij}e_i$ they are glued by $e_j^* = g_{ij}^{-1}e_i^*$.

Prerequisites: Definition 97, Definition 44.

Example.

For $K_{\mathbf{CP}^1}$, the dual bundle is $T^{1,0}\mathbf{CP}^1$. Since $dw = -z^{-2}dz$, the dual frame transforms as $\partial_w = -z^2\partial_z$. Evaluating the two dual frames gives 1 in either chart.

102. Order of a meromorphic function

A meromorphic function on a connected Riemann surface is locally a quotient of holomorphic functions with denominator not identically zero. For a nonzero meromorphic function f and a local coordinate t with $t(p) = 0$, write $f = t^m u$, where $m \in \mathbb{Z}$ and u is holomorphic near p .

102. Order of a meromorphic function (continued)

with $u(p) \neq 0$. The uniquely determined integer m is $\text{ord}_p(f)$. Positive and negative orders mean zeros and poles respectively; the identically zero function is excluded.

Prerequisites: Definition 61, Definition 40.

Example.

The rational function $f = z^3/(z - 2)$ on $\mathbf{CP}^1$ has orders 3 at 0 and -1 at 2. At infinity, $f = w^{-2}/(1 - 2w)$, so its order there is -2 . At every other point its order is zero.

103. Divisor on a compact Riemann surface

A divisor on a connected compact Riemann surface X is a finite formal sum $D = \sum_{p \in X} n_p [p]$ with integers n_p , all but finitely many zero. Divisors are added coefficientwise. It is effective if every $n_p \geq 0$, and $D \geq E$ means $D - E$ is effective.

Prerequisites: Definition 61.

Example.

The formal sum $D = 2[0] - [\infty]$ is a divisor on $\mathbf{CP}^1$, with support $\{0, \infty\}$. It is not effective because its infinity coefficient is negative. The divisor $2[0] + [2]$ is effective.

104. Principal divisor

For a nonzero global meromorphic function f on a connected compact Riemann surface X , its principal divisor is $\text{div}(f) = \sum_{p \in X} \text{ord}_p(f) [p]$. This sum has finite support. A divisor is principal if it arises in this way.

Prerequisites: Definition 102, Definition 103.

Example.

On $\mathbf{CP}^1$, the meromorphic function z has principal divisor $\text{div}(z) = [0] - [\infty]$. Its zero at 0 and pole at infinity both have order one; its divisor degree is zero.

105. Linear equivalence of divisors

Divisors D, E on a connected compact Riemann surface are linearly equivalent, written $D \sim E$, if $D - E = \text{div}(f)$ for some nonzero global meromorphic function f . The divisor class group is the group of divisors modulo its subgroup of principal divisors.

Prerequisites: Definition 103, Definition 104.

Example.

On $\mathbf{CP}^1$, $[0]$ and $[\infty]$ are linearly equivalent because their difference is $\text{div}(z)$. Also $-2[0]$ and $-2[\infty]$ differ by $\text{div}(z^{-2})$. Equality of divisors is stronger than linear equivalence.

106. Degree of a divisor

For a divisor $D = \sum_p n_p [p]$ on a compact Riemann surface, its degree is $\deg D = \sum_p n_p \in \mathbb{Z}$. Each complex point has residue field $\mathbb{C}$, so no additional residue-field weight occurs.

Prerequisites: Definition 103.

Example.

On $\mathbf{CP}^1$, $D = 3[0] - [2] - 2[\infty]$ has degree $3 - 1 - 2 = 0$, while $\operatorname{div}(dz) = -2[\infty]$ has degree -2 . The first is a function divisor; the second is a one-form's divisor.

107. The invertible sheaf $\mathcal{O}(D)$

Let $D = \sum_p n_p [p]$ be a divisor on a connected compact Riemann surface X . For open $U \subseteq X$, let $\mathcal{O}_X(D)(U)$ consist of meromorphic functions f on U satisfying $\operatorname{ord}_p(f) + n_p \geq 0$ for every $p \in U$, together with zero. On a component where f is identically zero the condition is automatic. Addition, multiplication by holomorphic functions, and restriction are those of functions. Near a point p isolated from the other points of the support of D , a coordinate t with $t(p) = 0$ gives $\mathcal{O}_X(D) = t^{-n_p} \mathcal{O}_X$.

Prerequisites: Definition 77, Definition 80, Definition 102, Definition 103.

Example.

For $D = [\infty]$ on $\mathbf{CP}^1$, the analytic sheaf $\mathcal{O}_X(D)$ has local frame 1 on U_0 and w^{-1} near infinity. A global section is a meromorphic function with at most a simple pole at infinity and no other poles; the global section space is $\mathbb{C}\langle 1, z \rangle$. As a section of this line bundle, 1 has a zero at infinity because $1 = w \cdot w^{-1}$.

108. Canonical bundle and canonical sheaf

For a complex n -manifold X , the canonical line bundle is $K_X = \bigwedge^n (T^{1,0}X)^*$. Its sheaf of holomorphic sections is the canonical sheaf. In holomorphic coordinates its frame is $dz^1 \wedge \cdots \wedge dz^n$, with transformation law

$$dw^1 \wedge \cdots \wedge dw^n = \det \left(\frac{\partial w^i}{\partial z^j} \right) dz^1 \wedge \cdots \wedge dz^n.$$

On a Riemann surface it is the holomorphic cotangent line bundle.

Prerequisites: Definition 62, Definition 44, Definition 97, Definition 99.

Example.

For $X = \mathbf{CP}^1$, the complex dimension is one, so $K_X = (T^{1,0}X)^*$. The local frames dz, dw satisfy $dw = -z^{-2}dz$. The meromorphic section dz has a second-order pole at infinity and no other zero or pole, giving $K_X \simeq \mathcal{O}(-2)$.

109. Divisor of a meromorphic section

A meromorphic section of a holomorphic line bundle on a connected compact Riemann surface is given in local holomorphic frames by $s = f_i e_i$, with f_i meromorphic and $f_i = g_{ij} f_j$ on overlaps. For a section not identically zero, define $\text{ord}_p(s) = \text{ord}_p(f_i)$ and $\text{div}(s) = \sum_p \text{ord}_p(s)[p]$. These orders are independent of the frame because transition functions are holomorphic and nowhere zero. When the bundle is K_X , this is a canonical divisor.

Prerequisites: Definition 99, Definition 102, Definition 103, Definition 108.

Example.

On $\mathbf{CP}^1$, $dz = -w^{-2}dw$ gives $\text{div}(dz) = -2[\infty]$. The different meromorphic section $dz/z = -dw/w$ has divisor $-[0] - [\infty]$. Their difference is the principal divisor of $1/z$, so they give the same canonical divisor class.

110. The line bundle $\mathcal{O}(n)$ on the projective line

On $\mathbf{CP}^1$, put $U_0 = \{Z_0 \neq 0\}$, $U_\infty = \{Z_1 \neq 0\}$, $z = Z_1/Z_0$, and $w = Z_0/Z_1$. For an integer n , the line bundle $\mathcal{O}(n)$ is obtained from trivial bundles on these charts by gluing frames with $e_\infty = z^n e_0$. The same notation denotes its sheaf of holomorphic sections. Global sections have components satisfying $f_0(z) = z^n f_\infty(1/z)$.

Prerequisites: Definition 99, Definition 63.

Example.

On $\mathbf{CP}^1$, $\mathcal{O}(n) = \mathcal{O}(n[\infty])$ has frames $e_0 = 1$ and $e_\infty = w^{-n}$, so $e_\infty = z^n e_0$. A global section for $n \geq 0$ is represented on U_0 by a polynomial of degree at most n . For $n=2$, the basis is $1, z, z^2$; for $n=-2$ there are no nonzero global holomorphic sections.

Metrics, connections and curvature

111. Hermitian metric on a line bundle

A Hermitian metric on a holomorphic line bundle L is a family of positive-definite Hermitian forms $h_x : L_x \times L_x \rightarrow \mathbb{C}$, linear in the first variable, such that $h_e(x) = h_x(e(x), e(x))$ is a smooth positive function for every local holomorphic frame e . Under $e_j = g_{ij} e_i$, these functions satisfy $h_{e_j} = |g_{ij}|^2 h_{e_i}$.

Prerequisites: Definition 97, Definition 99, Definition 57.

Example.

For $K_{\mathbf{CP}^1}$ induced from the fixed unit-sphere metric, put $h_z = \|dz\|^2 = (1 + |z|^2)^2/2$. Then $h_w = |z|^{-4} h_z = (1 + |w|^2)^2/2$. Both functions are positive and smooth, including $w=0$, and obey the frame transition law.

112. Complex connection

Let L be a holomorphic line bundle on X . Write $\mathcal{A}^k(L)(U)$ for the smooth sections of $(\bigwedge^k T_{\mathbb{R}}^*U \otimes_{\mathbb{R}} \mathbb{C}) \otimes_{\mathbb{C}} L|_U$. A complex connection is a complex-linear operation $\nabla : \mathcal{A}^0(L) \rightarrow \mathcal{A}^1(L)$ compatible with restriction and satisfying

$$\nabla(fs) = df \otimes s + f\nabla s$$

for every open U , smooth $f : U \rightarrow \mathbb{C}$, and smooth section s on U . In a local frame write $\nabla e = A_e \otimes e$.

Prerequisites: Definition 98, Definition 18, Definition 19.

Example.

In the holomorphic frame dz of $K_{\mathbb{CP}^1}$, take $A_z = 2\bar{z}(1 + |z|^2)^{-1}dz$. The connection acts by $\nabla(f dz) = (df + fA_z) \otimes dz$. The corresponding form in the other frame is $A_w = 2\bar{w}(1 + |w|^2)^{-1}dw$; it satisfies $A_w = A_z + g^{-1}dg$ for $g = -z^{-2}$.

113. Dolbeault operator on a holomorphic line bundle

For a holomorphic line bundle L , its Dolbeault operator is the complex-linear local operator from smooth sections to smooth L -valued $(0,1)$ -forms defined by $\bar{\partial}_L(fe) = (\bar{\partial}f) \otimes e$ in a holomorphic frame e , with f smooth. Holomorphic transition functions make these expressions agree on overlaps. The $(0,1)$ part of a connection is obtained by projecting its output to L -valued $(0,1)$ -forms.

Prerequisites: Definition 98, Definition 20.

Example.

On the z -chart of $\mathbb{CP}^1$, the smooth section $s = \bar{z} dz$ of K_X satisfies $\bar{\partial}_K s = d\bar{z} \otimes dz$, which is nonzero. The section $z^2 dz$ instead has $\bar{\partial}_K(z^2 dz) = 0$. A smooth section and a holomorphic section are thus distinguished explicitly.

114. Curvature of a line bundle connection

Extend a connection on a line bundle to bundle-valued forms by

$$\nabla(\alpha \otimes s) = d\alpha \otimes s + (-1)^k \alpha \wedge \nabla s, \quad \deg \alpha = k.$$

Its curvature is the global complex-valued two-form F_{∇} determined by $\nabla^2 s = F_{\nabla} \otimes s$. Fibre endomorphisms of a line are canonically scalars. If $\nabla e = A_e \otimes e$ in a local frame, then $F_{\nabla} = dA_e$, since $A_e \wedge A_e = 0$.

Prerequisites: Definition 112, Definition 66.

Example.

For the preceding connection on $K_{\mathbb{CP}^1}$, differentiation gives $F_K = dA_z = -2(1 + |z|^2)^{-2} dz \wedge d\bar{z}$. Hence $\nabla^2(f dz) = f F_K \otimes dz$. The expression in the w -chart is identical after coordinate substitution.

115. Chern curvature of the canonical line bundle

For a Riemann surface X and a smooth Hermitian metric h on K_X , the Chern curvature of the canonical line bundle is the curvature $F_{K_X, h}$ of the Chern connection of (K_X, h) . In a holomorphic coordinate z , let $h_z = h(dz, dz)$. Then

$$F_{K_X, h} = -\partial\bar{\partial} \log h_z.$$

This is a global complex-valued $(1, 1)$ -form; specifying the metric is part of the data.

Prerequisites: Definition 108, Definition 13.

Example.

For the canonical bundle of $\mathbf{CP}^1$ with the fixed sphere-induced metric, $F_{K, h} = -2(1 + |z|^2)^{-2} dz \wedge d\bar{z} = i\Omega$. Thus $\frac{i}{2\pi} \int_{\mathbf{CP}^1} F_{K, h} = -\frac{1}{2\pi} \int_{S^2} \Omega = -2$. This uses the identification of the sphere with $\mathbf{CP}^1$ and its outward orientation.

116. First Chern form

The first Chern form of a holomorphic line bundle L with smooth Hermitian metric h is the closed real $(1, 1)$ -form $c_1(L, h) = \frac{i}{2\pi} F_{L, h}$, where $F_{L, h}$ is the curvature with convention $\nabla^2 = F_{L, h}$.

Prerequisites: Definition 13.

Example.

For $K_{\mathbf{CP}^1}$, $c_1(K, h) = \frac{i}{2\pi} F_K = -\Omega/(2\pi)$ and its integral is -2 . For the dual tangent line bundle, the form is $+\Omega/(2\pi)$ and its integral is $+2$. The normalization and the choice of bundle both affect the sign.

Complexes and homological algebra

117. Additive functor

A functor T between additive categories is additive if $T(f + g) = T(f) + T(g)$ for every pair of objects A, B and every $f, g \in \text{Hom}(A, B)$. An additive category has abelian morphism groups, bilinear composition, a zero object, and finite biproducts.

Prerequisites: Definition 45, Definition 53.

Example.

The global-section functor on abelian sheaves over S^2 is additive. For sheaf maps $\phi, \psi : \mathcal{F} \rightarrow \mathcal{G}$ and a global section s , $\Gamma(\phi + \psi)(s) = \phi_X(s) + \psi_X(s)$. It also sends $\mathcal{F} \oplus \mathcal{G}$ to $\mathcal{F}(S^2) \oplus \mathcal{G}(S^2)$.

118. Exactness at a term

A sequence of homomorphisms of abelian groups $A \xrightarrow{f} B \xrightarrow{g} C$ is exact at B if $\text{im } f = \ker g$. The same definition uses categorical images and kernels in an abelian category.

118. Exactness at a term (continued)

Prerequisites: Definition 54, Definition 55.

Example.

On S^2 , use the constant integer sections. The sequence $\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \xrightarrow{\text{mod } 2} \mathbb{Z}/2\mathbb{Z}$ is exact at the middle group: its incoming image and outgoing kernel are both $2\mathbb{Z}$. The maps are induced by maps of the corresponding constant sheaves on S^2 .

119. Short exact sequence

A short exact sequence of abelian groups is a sequence $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ exact at every term: f is injective, g is surjective, and $\text{im } f = \ker g$. In an abelian category, injective and surjective are replaced by monomorphism and epimorphism.

Prerequisites: Definition 118.

Example.

For $p = 0 \in \mathbf{CP}^1$ and $i : \{p\} \hookrightarrow X$, the sequence $0 \rightarrow \mathcal{I}_p \rightarrow \mathcal{O}_X \xrightarrow{\text{ev}_p} i_*\mathbb{C} \rightarrow 0$ is short exact. At p it is $0 \rightarrow (z) \rightarrow \mathbb{C}\{z\} \rightarrow \mathbb{C} \rightarrow 0$; away from p it is $0 \rightarrow \mathcal{O}_{X,q} \xrightarrow{\text{Id}} \mathcal{O}_{X,q} \rightarrow 0 \rightarrow 0$.

120. Left exact functor

A covariant additive functor $T : \mathcal{A} \rightarrow \mathcal{B}$ between abelian categories is left exact if every short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ induces an exact sequence $0 \rightarrow T(A) \rightarrow T(B) \rightarrow T(C)$. No surjectivity onto $T(C)$ is required.

Prerequisites: Definition 117, Definition 119.

Example.

The exponential sequence on $\mathbf{CP}^1$, $0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_X \xrightarrow{f \mapsto e^{2\pi i f}} \mathcal{O}_X^\times \rightarrow 1$, is exact as a sequence of abelian sheaves. Global sections give $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{C} \rightarrow \mathbb{C}^\times$, which is exact. This illustrates preservation of the left end. Local logarithms establish sheaf surjectivity; left exactness alone would not imply surjectivity on global sections.

121. Cochain complex

A cochain complex of abelian groups is a family $(C^q, d^q)_{q \in \mathbb{Z}}$ with homomorphisms $d^q : C^q \rightarrow C^{q+1}$ such that $d^{q+1} \circ d^q = 0 : C^q \rightarrow C^{q+2}$ for every integer q . It is concentrated in nonnegative degrees if $C^q = 0$ for all $q < 0$; this is an additional condition, not part of the general definition.

Prerequisites: Definition 38.

Example.

The de Rham cochain complex of S^2 is $C^0 = C^\infty(S^2, \mathbb{R})$, $C^1 = \Omega^1(S^2)$, $C^2 = \Omega^2(S^2)$, and $C^q = 0$ otherwise, with differential d . The condition $d^{q+1}d^q = 0$ is the equality of mixed

121. Cochain complex (continued)

partials in local coordinates.

122. Resolution

A resolution of an object F in an abelian category is an exact sequence $0 \rightarrow F \rightarrow L^0 \xrightarrow{d^0} L^1 \xrightarrow{d^1} L^2 \rightarrow \dots$. Exactness includes the terms F and L^0 .

Prerequisites: Definition 118, Definition 121.

Example.

On S^2 , the augmented de Rham sequence $0 \rightarrow \mathbb{R} \rightarrow \mathcal{A}^0 \xrightarrow{d} \mathcal{A}^1 \xrightarrow{d} \mathcal{A}^2 \rightarrow 0$ is a resolution. Exactness is a stalkwise statement: near every point a small coordinate disc admits primitives for closed forms (Poincaré lemma). Global area forms need not have global primitives.

123. Injective object

An object I of a category is injective if for every monomorphism $j : A \rightarrow B$ and every morphism $u : A \rightarrow I$, there exists a morphism $v : B \rightarrow I$ such that $v \circ j = u$. The extension need not be unique.

Prerequisites: Definition 52.

Example.

At a fixed $p \in S^2$, the skyscraper $i_*\mathbb{Q}$ is injective in abelian sheaves. Indeed $\text{Hom}(\mathcal{F}, i_*\mathbb{Q}) \simeq \text{Hom}(\mathcal{F}_p, \mathbb{Q})$, the stalk functor is exact, and the divisible group $\mathbb{Q}$ is injective. Concretely the map $n\mathbb{Z} \rightarrow \mathbb{Q}$, $n \mapsto a$, extends to $\mathbb{Z} \rightarrow \mathbb{Q}$, $1 \mapsto a/n$. This example is in abelian sheaves, not in $\mathcal{O}_X$ -modules.

124. Injective resolution

An injective resolution of an object F in an abelian category is an exact sequence $0 \rightarrow F \rightarrow I^0 \rightarrow I^1 \rightarrow \dots$ with I^q injective for every $q \geq 0$. The object F itself need not be injective.

Prerequisites: Definition 122, Definition 123.

Example.

For $i : \{p\} \hookrightarrow S^2$, $0 \rightarrow i_*\mathbb{Z} \rightarrow i_*\mathbb{Q} \rightarrow i_*(\mathbb{Q}/\mathbb{Z}) \rightarrow 0$ is an injective resolution of $i_*\mathbb{Z}$, concentrated in resolution degrees 0 and 1. Both $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ are divisible, hence injective abelian groups. The differential is the quotient map, and its kernel is precisely $\mathbb{Z}$.

125. Cocycle

For a cochain complex $(C^\bullet, d)$ and an integer q , a q -cocycle is an element of $Z^q(C^\bullet) = \ker(d^q : C^q \rightarrow C^{q+1})$.

Prerequisites: Definition 121, Definition 54.

125. Cocycle (continued)

Example.

In the de Rham complex of S^2 , the area form Ω is a degree-two cocycle: it is in C^2 and d sends it to zero in $C^3 = 0$. Being a cocycle does not say it is a coboundary.

126. Coboundary

For a cochain complex $(C^\bullet, d)$ and an integer q , a q -coboundary is an element of $B^q(C^\bullet) = \text{im}(d^{q-1} : C^{q-1} \rightarrow C^q)$. Thus it has the form $d^{q-1}b$ with $b \in C^{q-1}$.

Prerequisites: Definition 121, Definition 55.

Example.

In the de Rham complex of S^2 , $dH \in C^1$ is a degree-one coboundary because H is a global element of C^0 . It is also a cocycle, since $d(dH) = 0$.

127. Cohomology of a complex

The q th cohomology of a cochain complex of abelian groups is

$$H^q(C^\bullet) = \frac{\ker(d^q : C^q \rightarrow C^{q+1})}{\text{im}(d^{q-1} : C^{q-1} \rightarrow C^q)}.$$

The relation $d^q d^{q-1} = 0$ makes the denominator a subgroup of the numerator. Two cocycles represent the same class precisely when their difference is a coboundary.

Prerequisites: Definition 125, Definition 126, Definition 56.

Example.

For the de Rham complex of S^2 , $H^2(C^\bullet) = \Omega^2(S^2)/d\Omega^1(S^2) \simeq \mathbb{R}$. The class of $\Omega/(4\pi)$ is normalized to have integral one; the class of a global form $d\alpha$ is zero.

128. Right derived functor

Let $T : \mathcal{A} \rightarrow \mathcal{B}$ be a covariant additive left exact functor between abelian categories, with $\mathcal{A}$ having enough injectives. Choose an injective resolution $0 \rightarrow F \xrightarrow{\epsilon_F} I^0 \rightarrow I^1 \rightarrow \dots$. For $q \geq 0$, define $R^q T(F) = H^q(T(I^\bullet))$. For a morphism $u : F \rightarrow G$ and an injective resolution $0 \rightarrow G \xrightarrow{\epsilon_G} J^0 \rightarrow \dots$, choose a cochain map $\tilde{u} : I^\bullet \rightarrow J^\bullet$ with $\tilde{u}^0 \epsilon_F = \epsilon_G u$, and put $R^q T(u) = H^q(T(\tilde{u}))$. The comparison theorem gives independence from these choices up to canonical isomorphism. The complex $T(I^\bullet)$ starts at $T(I^0)$ and does not include $T(F)$ as a term.

Prerequisites: Definition 120, Definition 124, Definition 127.

Example.

For $F = \Gamma(S^2, -)$, the right derived functors satisfy $R^0 F(\mathbb{R}) = \mathbb{R}$, $R^1 F(\mathbb{R}) = 0$, and $R^2 F(\mathbb{R}) = \mathbb{R}$. The last group is computed by the acyclic de Rham resolution and represented by $\Omega/(4\pi)$.

128. Right derived functor (continued)

Using an acyclic resolution is a theorem about derived functors, not their definition.

129. Acyclic complex

A cochain complex $C^\bullet$ is acyclic if $H^q(C^\bullet) = 0$ for every integer q , including $q = 0$.

Prerequisites: Definition 127.

Example.

Let $A = \Gamma(S^2, \mathbb{R}) = \mathbb{R}$. The two-term complex $C^0 = A \xrightarrow{\text{Id}} C^1 = A$, zero in other degrees, is acyclic: $H^0 = \ker \text{Id} = 0$ and $H^1 = A/\text{im Id} = 0$. Its individual terms are nevertheless nonzero.

130. Morphism of cochain complexes

A morphism of cochain complexes $f : (C^\bullet, d_C) \rightarrow (D^\bullet, d_D)$ is a family of homomorphisms $f^q : C^q \rightarrow D^q$ such that $d_D^q f^q = f^{q+1} d_C^q$ for every integer q .

Prerequisites: Definition 121.

Example.

Complex conjugation on $\mathbf{CP}^1$ is the smooth map $c([S : T]) = [\bar{S} : \bar{T}]$, corresponding on the sphere to $(X, Y, Z) \mapsto (X, -Y, Z)$. Pullback c^* is a map of de Rham complexes, because $d(c^*\alpha) = c^*(d\alpha)$. It sends the area form to $-\Omega$.

131. Enough injectives

A category has enough injectives if every object F admits a monomorphism $F \hookrightarrow I$ into some injective object I .

Prerequisites: Definition 123.

Example.

For any abelian sheaf $\mathcal{F}$ on S^2 , choose group embeddings $\mathcal{F}_p \hookrightarrow I_p$ into injective abelian groups. Sending a section to all its germs defines a monomorphism $\mathcal{F} \hookrightarrow \prod_{p \in S^2} i_{p*} I_p$. Each factor is injective by stalk adjunction, and a product of injective objects is injective here. This is a concrete construction on the fixed sphere, using the standard enough-injectives theorem for abelian groups.

132. Chain complex

A chain complex of abelian groups is a family $(C_q, \partial_q)_{q \in \mathbb{Z}}$ with homomorphisms $\partial_q : C_q \rightarrow C_{q-1}$ such that $\partial_{q-1} \partial_q = 0$ for every integer q . It is concentrated in nonnegative degrees if $C_q = 0$ for $q < 0$; then $\partial_0 : C_0 \rightarrow 0$ is zero.

Prerequisites: Definition 38.

132. Chain complex (continued)

Example.

Triangulate S^2 by the boundary K of a tetrahedron. Its simplicial chain complex has $C_2 = \mathbb{Z}^4$, $C_1 = \mathbb{Z}^6$, $C_0 = \mathbb{Z}^4$, with $\partial[i, j, k] = [j, k] - [i, k] + [i, j]$. All groups in degrees above two vanish, and $\partial^2 = 0$.

133. Cycle

A q -cycle in a chain complex $(C_\bullet, \partial)$ is an element of $\ker(\partial_q : C_q \rightarrow C_{q-1})$.

Prerequisites: Definition 132, Definition 54.

Example.

In the tetrahedral model of S^2 , $c = [1, 2, 3] - [0, 2, 3] + [0, 1, 3] - [0, 1, 2]$ is a two-cycle. Applying the displayed boundary formula cancels every oriented edge exactly twice with opposite signs.

134. Boundary

A q -boundary in a chain complex $(C_\bullet, \partial)$ is an element of $\text{im}(\partial_{q+1} : C_{q+1} \rightarrow C_q)$.

Prerequisites: Definition 132, Definition 55.

Example.

On the tetrahedrally triangulated S^2 , the chain $b = [1, 2] - [0, 2] + [0, 1]$ is the boundary $\partial[0, 1, 2]$. It is a one-cycle, since the endpoint contributions in ∂b cancel.

135. Homology

For a chain complex of abelian groups, the q th homology is

$$H_q(C_\bullet) = \frac{\ker(\partial_q : C_q \rightarrow C_{q-1})}{\text{im}(\partial_{q+1} : C_{q+1} \rightarrow C_q)}.$$

Since $\partial_q \partial_{q+1} = 0$, the denominator is a subgroup of the numerator. Two cycles represent the same class precisely when their difference is a boundary.

Prerequisites: Definition 133, Definition 134, Definition 56.

Example.

For the tetrahedral model of S^2 , the class of c from the cycle example generates $H_2(K; \mathbb{Z}) \simeq \mathbb{Z}$. It is not a boundary in this simplicial complex because $C_3(K) = 0$. The one-cycle bounding a face represents zero in $H_1(K; \mathbb{Z})$.

Computing cohomology

136. Global section functor

For a fixed topological space X , let $\text{Sh}(X; \mathbf{Ab})$ be the category of sheaves of abelian groups and their morphisms. The global section functor $\Gamma(X, -) : \text{Sh}(X; \mathbf{Ab}) \rightarrow \mathbf{Ab}$ sends $\mathcal{F}$ to $\mathcal{F}(X)$ and a sheaf morphism f to its component f_X .

Prerequisites: Definition 81, Definition 76, Definition 45.

Example.

For $X = \mathbf{CP}^1$, the functor $\Gamma(X, -)$ gives $\Gamma(X, \mathcal{O}_X) = \mathbb{C}$ and $\Gamma(X, \mathcal{K}_X) = 0$. It gives $\Gamma(X, i_*\mathbb{C}) = \mathbb{C}$ for a skyscraper at any fixed p . The inputs are different sheaves on the same space.

137. Sheaf cohomology

For a sheaf of abelian groups $\mathcal{F}$ on X , choose an injective resolution $0 \rightarrow \mathcal{F} \rightarrow I^0 \xrightarrow{d^0} I^1 \rightarrow \dots$. Set $D^r = \Gamma(X, d^r)$, $I^{-1}(X) = 0$, and $D^{-1} = 0$. For $q \geq 0$, its sheaf cohomology is

$$H^q(X, \mathcal{F}) = \frac{\ker(D^q : I^q(X) \rightarrow I^{q+1}(X))}{\text{im}(D^{q-1} : I^{q-1}(X) \rightarrow I^q(X))} = R^q\Gamma(X, \mathcal{F}).$$

The resolution is exact as a sequence of sheaves, before taking global sections.

Prerequisites: Definition 136, Definition 128.

Example.

For the real constant sheaf on S^2 , $H^2(S^2, \mathbb{R}) \simeq \mathbb{R}$, computed by the fine de Rham resolution and represented there by $\Omega/(4\pi)$. On the analytic $\mathbf{CP}^1$, $H^1(X, \mathcal{O}_X) = 0$ but $H^1(X, \mathcal{K}_X) \simeq \mathbb{C}$. The coefficient sheaf is essential.

138. Cohomology dimension and Euler characteristic

For an $\mathcal{O}_X$ -module sheaf E on a complex manifold, put $h^q(X, E) = \dim_{\mathbb{C}} H^q(X, E)$ whenever this dimension is finite. If it is finite for all $q \geq 0$ and zero for all sufficiently large q , define $\chi(X, E) = \sum_{q \geq 0} (-1)^q h^q(X, E)$.

Prerequisites: Definition 137, Definition 43.

Example.

On $\mathbf{CP}^1$, $h^0(\mathcal{O}(n)) = \max(n+1, 0)$ and $h^1(\mathcal{O}(n)) = \max(-n-1, 0)$, so $\chi(\mathcal{O}(n)) = n+1$. In particular $\chi(K) = -1$ while $\deg K = -2$; for the tangent line, $\chi(T^{1,0}X) = 3$ while its degree is 2. These are different invariants.

139. Support of a function

The support of a real- or complex-valued function f on a topological space X is $\text{supp } f = \overline{\{x \in X : f(x) \neq 0\}}$, where the closure is taken in X .

Prerequisites: Definition 29, Definition 7.

Example.

Use the smooth function $\eta(t) = e^{-1/t}$ for $t > 0$ and 0 for $t \leq 0$. On S^2 , $f(p) = \eta(Z(p))$ has support the closed northern hemisphere $\{Z \geq 0\}$: support includes the equator although f vanishes there.

140. Support of a sheaf endomorphism

For an endomorphism $h : \mathcal{F} \rightarrow \mathcal{F}$ of a sheaf of abelian groups on X , define $\text{supp } h = \overline{\{x \in X : h_x \neq 0\}}$ in X . Here $h_x : \mathcal{F}_x \rightarrow \mathcal{F}_x$ sends $[U, s]_x$ to $[U, h_U(s)]_x$, and $h_x \neq 0$ means it is not the zero homomorphism.

Prerequisites: Definition 76, Definition 73, Definition 29.

Example.

On the smooth function sheaf $\mathcal{A}^0$ of S^2 , multiplication m_{ρ_0} is a sheaf endomorphism. Its support is $\{Z \geq -1/2\}$, the closure of the points where the germ of ρ_0 is nonzero. At a boundary point the value may be zero while the germ is not the zero germ.

141. Fine sheaf

On a paracompact Hausdorff space X , a sheaf of abelian groups $\mathcal{F}$ is fine if for every locally finite open cover (U_i) there are sheaf endomorphisms $h_i : \mathcal{F} \rightarrow \mathcal{F}$ satisfying $\text{supp } h_i \subseteq U_i$ and $\sum_i h_i = \text{Id}_{\mathcal{F}}$. The sum is locally finite, because the supports are subordinate to the locally finite cover.

Prerequisites: Definition 140, Definition 30.

Example.

For the two-chart cover of S^2 , the endomorphisms m_{ρ_0} and m_{ρ_∞} of $\mathcal{A}^k$ have supports inside the corresponding charts and sum to the identity. For any locally finite cover, a subordinate smooth partition yields the same construction, proving that smooth differential-form sheaves are fine. One fixed partition alone illustrates, but does not exhaust, the quantifier over covers.

142. Fine resolution

On a paracompact Hausdorff space X , a fine resolution of a sheaf of abelian groups $\mathcal{F}$ is an exact sequence of sheaves $0 \rightarrow \mathcal{F} \rightarrow \mathcal{L}^0 \rightarrow \mathcal{L}^1 \rightarrow \cdots$ with every $\mathcal{L}^q$, $q \geq 0$, fine.

Prerequisites: Definition 122, Definition 141.

Example.

142. Fine resolution (continued)

The sequence $0 \rightarrow \mathbb{R} \rightarrow \mathcal{A}^0 \rightarrow \mathcal{A}^1 \rightarrow \mathcal{A}^2 \rightarrow 0$ on S^2 is a fine resolution: Poincaré lemma gives exactness, and subordinate partitions act by multiplication to make each $\mathcal{A}^k$ fine.

143. Acyclic sheaf

A sheaf $\mathcal{F}$ on a topological space X is acyclic on X if $H^q(X, \mathcal{F}) = 0$ for every integer $q > 0$. Vanishing of H^0 is not required.

Prerequisites: Definition 137.

Example.

The smooth function sheaf $\mathcal{A}^0$ on S^2 is acyclic for global sections: $H^q(S^2, \mathcal{A}^0) = 0$ for every $q > 0$, by the fine-sheaf acyclicity theorem on paracompact Hausdorff spaces. Its global sections are not zero; for instance it contains H and the constant 1.

144. Acyclic resolution

An acyclic resolution of a sheaf $\mathcal{F}$ on X is an exact sequence $0 \rightarrow \mathcal{F} \rightarrow \mathcal{L}^0 \rightarrow \mathcal{L}^1 \rightarrow \dots$ such that $H^q(X, \mathcal{L}^p) = 0$ for every $p \geq 0$ and every $q > 0$.

Prerequisites: Definition 143, Definition 122.

Example.

The fine de Rham resolution on S^2 is also an acyclic resolution, since its terms have no higher sheaf cohomology. Taking global sections therefore computes $H^q(S^2, \mathbb{R})$. No assertion that each term is an injective sheaf is needed.

145. Čech cochain

Let $\mathfrak{U} = (U_i)_{i \in I}$ be an open cover of X , with a total order on I , and let $\mathcal{F}$ be a sheaf of abelian groups. Put $U_{i_0 \dots i_p} = \bigcap_{a=0}^p U_{i_a}$. For $p \geq 0$, the group of Čech cochains is

$$\check{C}^p(\mathfrak{U}, \mathcal{F}) = \prod_{i_0 < \dots < i_p} \mathcal{F}(U_{i_0 \dots i_p}).$$

A cochain specifies one section on every such intersection. Addition is componentwise, and $\mathcal{F}(\emptyset) = 0$. The product imposes no finite-support condition.

Prerequisites: Definition 23, Definition 69.

Example.

For $\mathfrak{U} = (U_0, U_\infty)$ on $\mathbf{CP}^1$ and $\mathcal{K}_X$, a zero-cochain is $(a(z)dz, b(w)dw)$ with a, b entire. A one-cochain is $f(z)dz$ with f holomorphic on $\mathbb{C}^\times$. Thus the overlap has its own sections; it is not just the intersection of two vector spaces of global sections.

146. Čech differential / Coboundary operator

For the ordered open cover $\mathfrak{U}$ and sheaf $\mathcal{F}$, the Čech differential $\delta^p : \check{C}^p(\mathfrak{U}, \mathcal{F}) \rightarrow \check{C}^{p+1}(\mathfrak{U}, \mathcal{F})$ is

$$(\delta^p c)_{i_0 \dots i_{p+1}} = \sum_{a=0}^{p+1} (-1)^a c_{i_0 \dots \widehat{i_a} \dots i_{p+1}}|_{U_{i_0 \dots i_{p+1}}}.$$

The hat denotes omission, and each term is restricted to the entire displayed intersection. Put $\check{C}^{-1} = 0$ and $\delta^{-1} = 0$.

Prerequisites: Definition 145.

Example.

On the two standard charts of $\mathbf{CP}^1$, using the standard alternating convention $(\delta s)_{0\infty} = s_\infty - s_0$, the preceding zero-cochain maps, in the dz frame, to $(-z^{-2}b(1/z) - a(z))dz$. With two members there are no increasing triple indices, so the next alternating differential is zero.

147. Čech cohomology of a cover

For an ordered open cover $\mathfrak{U}$ of X and a sheaf $\mathcal{F}$, the Čech cohomology of the cover is

$$\check{H}^p(\mathfrak{U}, \mathcal{F}) = \frac{\ker(\delta^p : \check{C}^p \rightarrow \check{C}^{p+1})}{\text{im}(\delta^{p-1} : \check{C}^{p-1} \rightarrow \check{C}^p)}, \quad p \geq 0,$$

where all cochains are for $(\mathfrak{U}, \mathcal{F})$ and $\check{C}^{-1} = 0$.

Prerequisites: Definition 146, Definition 127.

Example.

For the analytic two-chart cover of $\mathbf{CP}^1$, $\check{H}^1(\mathfrak{U}, \mathcal{K}_X)$ is $\mathcal{O}(\mathbb{C}^\times)/(\mathcal{O}(\mathbb{C}) + z^{-2}\mathcal{O}(\mathbb{C})|_{w=1/z})$, times dz . Laurent expansion leaves precisely the coefficient of z^{-1} , so the group is $\mathbb{C}\langle [dz/z] \rangle$. Positive and sufficiently negative Laurent tails extend to the respective charts.

148. Čech cohomology of a space

For a sheaf $\mathcal{F}$ on X and $p \geq 0$, the Čech cohomology of the space is $\check{H}^p(X, \mathcal{F}) = \varinjlim_{\mathfrak{U}} \check{H}^p(\mathfrak{U}, \mathcal{F})$ over open covers ordered by refinement. A refinement $\mathfrak{V}$ of $\mathfrak{U}$ induces a map from the cohomology of $\mathfrak{U}$ to that of $\mathfrak{V}$, by restriction; its cohomology map is independent of the refinement assignment. Concretely, pairs $(\mathfrak{U}, \alpha)$ and $(\mathfrak{V}, \beta)$ are equivalent if their images agree on some common refinement. Addition is performed after passing to a common refinement.

Prerequisites: Definition 147, Definition 31.

Example.

Take the open-star good cover of the tetrahedral S^2 from the common conventions. Its degree-two constant-real Čech cocycle assigning 1 to the face (1,2,3) and 0 to the other faces gives the nonzero orientation class, after passing through refinement maps to full Čech cohomology.

148. Čech cohomology of a space (continued)

Its value on the oriented fundamental cycle is 1, so it cannot be a coboundary. Good-cover comparison identifies this with the generator of $H^2(S^2; \mathbb{R})$.

149. $\mathcal{F}$ -acyclic cover

An open cover (U_i) of X is $\mathcal{F}$ -acyclic if every nonempty finite intersection $W = U_{i_0} \cap \cdots \cap U_{i_p}$, including $p = 0$, satisfies $H^q(W, \mathcal{F}|_W) = 0$ for every $q > 0$.

Prerequisites: Definition 143, Definition 23.

Example.

On analytic $\mathbf{CP}^1$, $U_0 \simeq \mathbb{C}$, $U_\infty \simeq \mathbb{C}$, and their overlap $\mathbb{C}^\times$ are Stein. The restrictions of the locally free sheaf $\mathcal{K}_X$ have vanishing higher cohomology by Cartan's theorem B. Consequently this cover is $\mathcal{K}_X$ -acyclic and its Čech calculation computes sheaf cohomology. The Stein acyclicity theorem is an explicitly used theorem, not a property of every open cover.

150. Standard simplex

For $q \geq 0$, the standard q -simplex is $\Delta^q = \{(t_0, \dots, t_q) \in \mathbb{R}^{q+1} : t_i \geq 0 \text{ for every } i, \sum_{i=0}^q t_i = 1\}$, with the subspace topology.

Prerequisites: Definition 43.

Example.

For the triangle used to parametrize a face of the tetrahedral S^2 , $\Delta^2 = \{(t_0, t_1, t_2) \in \mathbb{R}^3 : t_i \geq 0, t_0 + t_1 + t_2 = 1\}$. Its barycentre is $(1/3, 1/3, 1/3)$, and its three vertices are the coordinate vectors. This is the parameter simplex, not the spherical image itself.

151. Singular simplex

A singular q -simplex of a topological space X is a continuous map $\sigma : \Delta^q \rightarrow X$, where $q \geq 0$. The map need not be injective.

Prerequisites: Definition 150, Definition 25.

Example.

Let v_0, v_1, v_2 be three vertices of the inscribed tetrahedron. The map $\sigma(t) = (\sum_{j=0}^2 t_j v_j) / \|\sum_{j=0}^2 t_j v_j\|$ from Δ^2 to S^2 is a singular two-simplex. Its denominator is nonzero because a tetrahedral face lies in a plane not through the origin.

152. Singular chain group

For $q \geq 0$, $C_q(X; \mathbb{R})$ is the real vector space with basis all singular q -simplices of X . Its elements are finite sums $\sum_{k=1}^N a_k \sigma_k$ with $a_k \in \mathbb{R}$. Set $C_q = 0$ for $q < 0$.

Prerequisites: Definition 151, Definition 32.

152. Singular chain group (continued)

Example.

The singular chain group $C_2(S^2; \mathbb{Z})$ consists of finite formal integer combinations of continuous maps $\Delta^2 \rightarrow S^2$. If σ_{012} and σ_{013} are radial face maps, $2\sigma_{012} - \sigma_{013}$ is an element. The coefficients combine maps formally; they do not add points of S^2 .

153. Singular boundary operator

For $q > 0$, let $\epsilon_i : \Delta^{q-1} \rightarrow \Delta^q$ insert zero into coordinate i . The singular boundary is defined on a simplex by $\partial_q \sigma = \sum_{i=0}^q (-1)^i \sigma \circ \epsilon_i$ and extended linearly to $C_q(X; \mathbb{R})$. Set $\partial_0 : C_0(X; \mathbb{R}) \rightarrow 0$ equal to zero.

Prerequisites: Definition 152.

Example.

For the radial face map $\sigma_{012} : \Delta^2 \rightarrow S^2$, its singular boundary is $\partial \sigma_{012} = \sigma_{12} - \sigma_{02} + \sigma_{01}$, where the edge maps are restrictions along the ordered face inclusions. Applying the endpoint boundary gives zero.

154. Singular homology

For $q \geq 0$, singular homology with real coefficients is $H_q^{\text{sing}}(X; \mathbb{R}) = \ker \partial_q / \text{im } \partial_{q+1}$ for the singular chain complex, with $C_{-1} = 0$ and $\partial_0 = 0$.

Prerequisites: Definition 153, Definition 135.

Example.

The alternating sum of the four radial face maps is a singular two-cycle on S^2 . Its homology class generates $H_2(S^2; \mathbb{R}) \simeq \mathbb{R}$. Unlike the simplicial complex, the singular chain complex does have three-chains; nontriviality here follows from the simplicial-to-singular comparison theorem (or the degree pairing), not from an assertion that $C_3^{\text{sing}} = 0$.

155. Singular cochain

A real singular q -cochain is a real-linear map $C_q(X; \mathbb{R}) \rightarrow \mathbb{R}$. Thus $C^q(X; \mathbb{R}) = \text{Hom}_{\mathbb{R}}(C_q(X; \mathbb{R}), \mathbb{R})$, with no finite-support condition on the cochain.

Prerequisites: Definition 152, Definition 38.

Example.

Fix $N \in S^2$. A singular real zero-cochain can be defined by $a(p) = 1$ if $p=N$ and 0 otherwise, and extended linearly to finite zero-chains. Singular cochains are arbitrary homomorphisms from chain groups; this example need not be a continuous function of p .

156. Singular coboundary operator

The singular coboundary $\delta^q : C^q(X; \mathbb{R}) \rightarrow C^{q+1}(X; \mathbb{R})$ is defined by $\delta^q(c) = c \circ \partial_{q+1}$. For every $(q+1)$ -chain z , $(\delta^q c)(z) = c(\partial_{q+1} z)$.

Prerequisites: Definition 155, Definition 153.

Example.

For that zero-cochain a and a path $\gamma : [0, 1] \rightarrow S^2$ from S to N , $(\delta a)(\gamma) = a(\partial \gamma) = a(N) - a(S) = 1$. Thus the cochain differential is dual to the chain boundary, with the endpoint signs visible.

157. Singular cohomology

For $q \geq 0$, real singular cohomology is $H_{\text{sing}}^q(X; \mathbb{R}) = \ker \delta^q / \text{im } \delta^{q-1}$ for the dual of the real singular chain complex, where $\delta^q c = c \circ \partial_{q+1}$, $C^{-1} = 0$, and $\delta^{-1} = 0$.

Prerequisites: Definition 156, Definition 127.

Example.

For the sphere, $H^0(S^2; \mathbb{R}) = \mathbb{R}$, $H^1(S^2; \mathbb{R}) = 0$, and $H^2(S^2; \mathbb{R}) = \mathbb{R}$. The degree-two class is normalized to evaluate to 1 on the oriented fundamental cycle. Under the de Rham comparison theorem it corresponds to $[\Omega/(4\pi)]$.

158. Abstract simplicial complex

An abstract simplicial complex K on a vertex set is a collection of finite subsets such that $\sigma \in K$ and $\tau \subseteq \sigma$ imply $\tau \in K$. Its members are simplices; a simplex with $q+1$ vertices has dimension q .

Example.

Let K consist of all proper subsets of $\{0, 1, 2, 3\}$. It is an abstract simplicial complex: subsets of proper subsets are proper. Its realization is the boundary of a tetrahedron and hence homeomorphic to S^2 . The full four-vertex subset is deliberately absent.

159. Simplicial chain complex

Give the vertices of a simplicial complex K a total order. For $q \geq 0$, $C_q(K; \mathbb{R})$ is the vector space of finite linear combinations of simplices $[v_0, \dots, v_q]$ with $v_0 < \dots < v_q$. For $q > 0$, define

$$\partial_q[v_0, \dots, v_q] = \sum_{i=0}^q (-1)^i [v_0, \dots, \widehat{v_i}, \dots, v_q]$$

and extend linearly. Put $C_{-1} = 0$ and $\partial_0 = 0$. This gives the simplicial chain complex.

Prerequisites: Definition 158, Definition 132.

Example.

For this K modeling S^2 , choose increasing orientations on simplices. The chain groups are free

159. Simplicial chain complex (continued)

on four vertices, six edges and four faces. For example $\partial[0, 1, 2] = [1, 2] - [0, 2] + [0, 1]$ and $\partial[0, 1] = [1] - [0]$.

160. Simplicial homology

For $q \geq 0$, real simplicial homology is $H_q(K; \mathbb{R}) = \ker \partial_q / \text{im } \partial_{q+1}$ for the ordered simplicial chain complex $C_\bullet(K; \mathbb{R})$, with $C_{-1} = 0$ and $\partial_0 = 0$.

Prerequisites: Definition 159, Definition 135.

Example.

For the tetrahedral boundary modeling S^2 , the simplicial homology groups are $H_0(K; \mathbb{R}) = \mathbb{R}$, $H_1(K; \mathbb{R}) = 0$, $H_2(K; \mathbb{R}) = \mathbb{R}$. The oriented sum of the four faces is the degree-two generator; a face boundary is a zero class in degree one.

161. Simplicial cohomology

For a simplicial complex K , put $C^q(K; \mathbb{R}) = \text{Hom}_{\mathbb{R}}(C_q(K; \mathbb{R}), \mathbb{R})$ and $\delta^q c = c \circ \partial_{q+1}$. Its simplicial cohomology is $H^q(K; \mathbb{R}) = \ker \delta^q / \text{im } \delta^{q-1}$, with cochain groups zero in negative degrees.

Prerequisites: Definition 159, Definition 127.

Example.

For K modeling S^2 , define a two-cochain a by $a([1, 2, 3]) = 1$ and $a=0$ on the other increasing faces. It is a cocycle since $C^3(K) = 0$. It evaluates to 1 on the fundamental cycle, whereas every coboundary evaluates to zero, so its class is nonzero in $H^2(K; \mathbb{R})$.

162. Nerve of a cover

The nerve $N(\mathfrak{U})$ of an open cover $(U_i)_{i \in I}$ is the abstract simplicial complex consisting of the empty simplex and all nonempty finite subsets $J \subseteq I$ with $\bigcap_{i \in J} U_i \neq \emptyset$. Its vertices are the indices i with $U_i \neq \emptyset$.

Prerequisites: Definition 158, Definition 23.

Example.

Cover the tetrahedral S^2 by the four open vertex stars $U_0, \dots, U_3$. A collection of stars has nonempty intersection exactly when its vertex set is a face of K . Thus its nerve is K itself: all edges and triangular faces occur, but the fourfold intersection is empty.

163. Contractible space

A space W is contractible if there exist $x_0 \in W$ and a continuous map $H : W \times [0, 1] \rightarrow W$ such that $H(x, 0) = x$ and $H(x, 1) = x_0$ for every $x \in W$.

163. Contractible space (continued)

Prerequisites: Definition 25.

Example.

The coordinate chart $U_0 \subset \mathbf{CP}^1$ is contractible: in its coordinate z , $F(z, t) = (1 - t)z$ contracts it to 0. This does not contract all of $\mathbf{CP}^1$, since the formula does not define that homotopy at infinity when $t=1$. The whole sphere is not contractible.

164. Good cover

A good cover of X is an open cover for which every nonempty finite intersection of members is contractible. This includes each individual member.

Prerequisites: Definition 163, Definition 23.

Example.

The four open vertex stars of the tetrahedral sphere form a good cover. Each nonempty finite intersection is an open star of a simplex and contracts to that simplex's barycentre by linear interpolation in barycentric coordinates, followed by the fixed homeomorphism to S^2 . The standard two-chart cover is not good: its overlap is $\mathbb{C}^\times$, which is not contractible.

Projective geometry and pure states

165. Projective hypersurface

For $n \geq 1$ and a nonzero homogeneous polynomial $F \in \mathbb{C}[Z_0, \dots, Z_n]$ of positive degree, its projective zero locus is $V(F) = \{[Z] \in \mathbf{CP}^n : F(Z) = 0\}$. If F is square-free, this reduced hypersurface has degree $\deg F$, and is smooth exactly when its partial derivatives do not vanish simultaneously at any point of $V(F)$. Using F^2 instead gives the same set but a different nonreduced hypersurface.

Prerequisites: Definition 63, Definition 33.

Example.

In $\mathbf{CP}^1$ with coordinates $[S:T]$, the homogeneous polynomial T of degree one cuts out the projective hypersurface $\{T = 0\} = \{[1 : 0]\}$. The polynomial T^2 has the same point set, but scheme-theoretically gives a double point; the set and its multiplicity data must be distinguished.

166. Segre embedding

For $m, n \geq 0$, the Segre embedding is

$$\sigma : \mathbf{CP}^m \times \mathbf{CP}^n \longrightarrow \mathbf{CP}^{(m+1)(n+1)-1}, \quad ([a], [b]) \longmapsto [a_i b_j]_{0 \leq i \leq m, 0 \leq j \leq n}.$$

Changing either representative multiplies every output coordinate by a common nonzero scalar. Its image is the set of projective classes of nonzero rank-one matrices, equivalently those whose 2×2 minors all vanish.

166. Segre embedding (continued)

Prerequisites: Definition 63, Definition 44.

Example.

The Segre map for two copies of $\mathbf{CP}^1$ sends $([a : b], [c : d])$ to $[ac : ad : bc : bd] \in \mathbf{CP}^3$. Restricting the second factor to $[1:0]$ gives the explicit copy $[a : b] \mapsto [a : 0 : b : 0]$ of $\mathbf{CP}^1$. General image coordinates satisfy $x_0x_3 - x_1x_2 = 0$.

167. Pure quantum state

For a finite-dimensional complex Hilbert space H , a pure state is a one-dimensional complex subspace of H , hence a point of $\mathbf{P}(H)$. Unit vectors ψ and $e^{i\theta}\psi$ represent the same state. Equivalently, with the Hermitian product linear in its first argument, this state is the rank-one orthogonal projector $v \mapsto h(v, \psi)\psi$.

Prerequisites: Definition 63, Definition 57.

Example.

For a two-level system, rays $[a : b] \in \mathbf{CP}^1$ represent pure states. On $a \neq 0$ choose $\psi_z = (1, z)/\sqrt{1 + |z|^2}$. Its Bloch vector is $(2\Re z, 2\Im z, 1 - |z|^2)/(1 + |z|^2) \in S^2$. Multiplying ψ_z by a unit complex phase changes neither its ray nor this sphere point.

168. Product state and pure-state entanglement

For finite-dimensional complex Hilbert spaces H_A, H_B , a pure state $[\psi] \in \mathbf{P}(H_A \otimes_{\mathbb{C}} H_B)$ is a product state if $[\psi] = [a \otimes b]$ for some nonzero $a \in H_A, b \in H_B$. A pure state which is not a product state is entangled. The product states form the image of the Segre embedding.

Prerequisites: Definition 166, Definition 167, Definition 44.

Example.

The family $[a : b] \in \mathbf{CP}^1 \mapsto [a : 0 : 0 : b] \in \mathbf{CP}^3$ represents bipartite pure states $a|00\rangle + b|11\rangle$. Its coefficient matrix is $\text{diag}(a, b)$. It is a product state exactly when $ab=0$; otherwise it has rank two and is entangled. Thus one fixed projective-line family contains both product endpoints and entangled interior points.

Propositions

Snake lemma and Mayer–Vietoris sequence

For a commutative diagram of abelian groups with short exact rows and vertical maps $a : A' \rightarrow A, b : B' \rightarrow B, c : C' \rightarrow C$, there is an exact sequence

$$0 \rightarrow \ker a \rightarrow \ker b \rightarrow \ker c \rightarrow \text{coker } a \rightarrow \text{coker } b \rightarrow \text{coker } c \rightarrow 0.$$

A short exact sequence of cochain complexes induces a long exact sequence in cohomology. For an open cover $M = U \cup V$, applying this to the de Rham complexes and the restriction

difference gives the Mayer–Vietoris sequence.

Example.

For the two charts of $\mathbf{CP}^1$, the connecting map sends $[dz/z] \in H_{\text{dR}}^1(U_0 \cap U_\infty; \mathbb{C})$ to a degree-two class. With lift difference $B_0 - B_\infty = dz/z$, Stokes gives $\int_{\mathbf{CP}^1} \delta[dz/z] = \oint_{|z|=1} dz/z = 2\pi i$.

Poincaré lemma and fine resolution

On a Hausdorff second-countable smooth manifold, the sequence $0 \rightarrow \underline{\mathbb{R}} \rightarrow \mathcal{A}^0 \xrightarrow{d} \mathcal{A}^1 \rightarrow \dots$ is an exact sequence of sheaves. Local exactness in positive degrees follows from the Poincaré lemma; in degree zero, functions with zero differential are locally constant. Smooth partitions of unity make every $\mathcal{A}^q$ fine.

Example.

The sequence $0 \rightarrow \underline{\mathbb{R}} \rightarrow \mathcal{A}^0 \rightarrow \mathcal{A}^1 \rightarrow \mathcal{A}^2 \rightarrow 0$ on S^2 is a fine resolution: Poincaré lemma gives exactness, and subordinate partitions act by multiplication to make each $\mathcal{A}^k$ fine.

Acyclic resolutions

The category of sheaves of abelian groups on a topological space X is abelian and has enough injectives. The functor $\Gamma(X, -)$ is left exact. Injective resolutions define sheaf cohomology independently of the resolution, up to canonical isomorphism, and $H^0(X, \mathcal{F}) = \mathcal{F}(X)$. If $0 \rightarrow \mathcal{F} \rightarrow \mathcal{L}^0 \rightarrow \mathcal{L}^1 \rightarrow \dots$ is exact and $H^q(X, \mathcal{L}^p) = 0$ for all $p \geq 0, q > 0$, then $H^q(\Gamma(X, \mathcal{L}^\bullet)) \simeq H^q(X, \mathcal{F})$. Fine sheaves on paracompact Hausdorff spaces are acyclic.

Example.

The fine de Rham resolution on S^2 is also an acyclic resolution, since its terms have no higher sheaf cohomology. Taking global sections therefore computes $H^q(S^2, \underline{\mathbb{R}})$. No assertion that each term is an injective sheaf is needed.

Leray comparison theorem

If $\mathfrak{U}$ is an $\mathcal{F}$ -acyclic open cover of X , the natural comparison map gives $\check{H}^p(\mathfrak{U}, \mathcal{F}) \simeq H^p(X, \mathcal{F})$ for every $p \geq 0$.

Example.

On analytic $\mathbf{CP}^1$, $U_0 \simeq \mathbb{C}$, $U_\infty \simeq \mathbb{C}$, and their overlap $\mathbb{C}^\times$ are Stein. The restrictions of the locally free sheaf $\mathcal{K}_X$ have vanishing higher cohomology by Cartan’s theorem B. Consequently this cover is $\mathcal{K}_X$ -acyclic and its Čech calculation computes sheaf cohomology. The Stein acyclicity theorem is an explicitly used theorem, not a property of every open cover.

Nerve comparison and de Rham theorem

If all nonempty finite intersections of an open cover are connected, its Čech cochain complex with coefficients in $\underline{\mathbb{R}}$ is the real simplicial cochain complex of its nerve. For a good cover of

a Hausdorff second-countable smooth manifold, these groups compute the cohomology of the manifold. Integration of forms over smooth singular simplices induces the de Rham isomorphism $H_{\text{dR}}^q(M) \simeq H_{\text{sing}}^q(M; \mathbb{R})$ for all $q \geq 0$.

Example.

The four open vertex stars of the tetrahedral sphere form a good cover. Each nonempty finite intersection is an open star of a simplex and contracts to that simplex's barycentre by linear interpolation in barycentric coordinates, followed by the fixed homeomorphism to S^2 . The standard two-chart cover is not good: its overlap is $\mathbb{C}^\times$, which is not contractible. For the sphere, $H^0(S^2; \mathbb{R}) = \mathbb{R}$, $H^1(S^2; \mathbb{R}) = 0$, and $H^2(S^2; \mathbb{R}) = \mathbb{R}$. The degree-two class is normalized to evaluate to 1 on the oriented fundamental cycle. Under the de Rham comparison theorem it corresponds to $[\Omega/(4\pi)]$.

Holomorphic germs and completion

For a local coordinate t centred at x on a Riemann surface, Taylor expansion gives $\mathcal{O}_{X,x} \simeq \mathbb{C}\{t\}$, with $\mathfrak{m}_x$ corresponding to (t) and $\widehat{\mathcal{O}}_{X,x} \simeq \mathbb{C}[[t]]$. Holomorphic germs are determined by their convergent Taylor series; smooth germs need not be determined by their Taylor series.

Example.

At $0 \in \mathbf{CP}^1$, the compatible truncations $1 + z + \cdots + z^k$ modulo z^{k+1} determine the element $\sum_{n \geq 0} z^n$ in $\widehat{\mathcal{O}}_{X,0} \simeq \mathbb{C}[[z]]$. The truncations of $\sum n! z^n$ also give an element of this completion even though no convergent germ has those coefficients.

Metric independence and degree

If two Hermitian metrics on a holomorphic line bundle satisfy $h' = e^\psi h$ for a global smooth real function ψ , then $F_{h'} - F_h = d(\partial\psi)$. Their first Chern forms differ by a real exact form. On a connected compact Riemann surface, for a nonzero meromorphic section s ,

$$\frac{i}{2\pi} \int_X F_{L,h} = \sum_{p \in X} \text{ord}_p(s).$$

Example.

For $K_{\mathbf{CP}^1}$, $c_1(K, h) = \frac{i}{2\pi} F_K = -\Omega/(2\pi)$ and its integral is -2 . For the dual tangent line bundle, the form is $+\Omega/(2\pi)$ and its integral is $+2$. The normalization and the choice of bundle both affect the sign.

Canonical bundle of the projective line

On $\mathbf{CP}^1$, the coordinate relation $w = 1/z$ gives $dw = -z^{-2}dz$, hence $\text{div}(dz) = -2[\infty]$, $K_{\mathbf{CP}^1} \simeq \mathcal{O}(-2)$, and $\frac{i}{2\pi} \int_{\mathbf{CP}^1} F_{K,h} = -2$.

Example.

For the canonical bundle of $\mathbf{CP}^1$ with the fixed sphere-induced metric, $F_{K,h} = -2(1 + |z|^2)^{-2}dz \wedge d\bar{z} = i\Omega$. Thus $\frac{i}{2\pi} \int_{\mathbf{CP}^1} F_{K,h} = -\frac{1}{2\pi} \int_{S^2} \Omega = -2$. This uses the identification of the sphere with $\mathbf{CP}^1$ and its outward orientation.

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